

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287099

Kind Code

A1

Publication Date

September 11, 2025

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PORTABLE INFORMATION TERMINAL AND IMAGE MANAGEMENT METHOD

Abstract

The problem is to prevent external leakage of images captured by a camera when using a camera mounted on a portable information terminal for imaging of image processing targets. The solution means that the security mode deletes the code image Ic generated by the rear camera **33** from the volatile memory **61** without storing it in the non-volatile memory **62** (step **S604** in FIG. **6D**). In the security mode, instead of this operation, a step of forcibly deleting the code image Ic stored in the non-volatile memory **62** from the non-volatile memory **62** may be executed. With this configuration, by selecting the security mode, the code image Ic generated by the rear camera **33** is deleted, so there is no leakage to the outside.

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Appl. No.: 19/064072
Filed: February 26, 2025

Foreign Application Priority Data

JP 2024-034090 Mar. 06, 2024

Publication Classification

Int. Cl.: H04N23/667 (20230101); G06F1/16 (20060101); H04M1/72463 (20210101); H04N1/21 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims foreign priority based on Japanese Patent Application No. 2024-034090, filed Mar. 6, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Technical Field

[0002] The present disclosure relates to a technology for reading codes attached to a workpiece.

2. Description of the Related Art

[0003] Japanese Patent Publication No. 2022-144178 discloses an information reading device that reads codes such as barcodes and QR codes (registered trademark).

[0004] This information reading device acquires images through a camera unit provided at the upper end of the housing and reads the codes by decoding the codes contained in the images.

[0005] There is a need to easily perform reading of codes as described above and image processing such as OCR (Optical Character Recognition/Reader) using portable information terminals represented by smartphones. In response to this, it is possible to meet such needs by using the camera mounted on the portable information terminal for capturing images of the image processing target. However, this camera can also be used for taking photographs of people and landscapes, and the images captured by the camera may contain information other than the image processing target. Therefore, in areas dealing with confidential information, it has been difficult to use the camera mounted on such portable information terminals for capturing images of the image processing target.

[0006] The present disclosure has been made in consideration of the above problems, and aims to provide a technology that can prevent external leakage of images captured by a camera used for imaging of image processing targets mounted on a portable information terminal.

SUMMARY OF THE INVENTION

[0007] The portable information terminal according to the present disclosure includes a camera that captures a workpiece and generates a captured image, a reading unit that executes image processing based on the captured image to acquire information contained in the workpiece, a first memory unit that is a volatile memory unit and temporarily stores the captured image, a second memory unit that is a non-volatile memory unit and stores settings of the image processing and is capable of storing the captured image, a control unit that causes the camera to generate the captured image and store the captured image in the first memory unit, and causes the reading unit to execute the image processing based on the captured image in the first memory, a mode selection unit that selects any of a plurality of modes including a first mode where the captured image is stored in the second memory unit and a second mode where the captured image is deleted from the first memory unit without the captured image being not stored in the second memory unit or the captured image is forcibly deleted from the second memory unit after being stored in the second memory.

[0008] The image management method according to the present disclosure is an image management method for managing a first memory unit or a second memory unit of a captured image generated by a camera that captures a workpiece, comprising: a step of selecting any of a plurality of modes including a first mode where the captured image generated by the camera is stored in the second memory unit, and a second mode where the captured image is deleted from the first memory unit without being stored in the second memory unit, or the captured image is forcibly deleted from the second memory unit after being stored in the second memory unit; and a step of

controlling the camera to generate the captured image and storing the captured image in the first memory unit, and executing image processing based on the captured image in the first memory unit to acquire information included in the workpiece, wherein the first memory unit is a volatile memory unit that temporarily stores the captured image, and the second memory unit is a non-volatile memory unit that is capable of storing the captured image and stores settings of the image processing.

[0009] In the present disclosure (portable information terminal and image management method) configured as such, by selecting the second mode, the captured image generated by the camera is deleted, preventing the captured image from leaking externally. Thus, when using the camera mounted on the portable information terminal for capturing the image processing target, it becomes possible to prevent external leakage of the images captured by the camera.

[0010] According to the above disclosure, it is possible to prevent external leakage of images captured by the camera when using the camera mounted on a portable information terminal for imaging of image processing targets.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1A is a front view showing the external configuration of the portable information terminal according to the present disclosure.

[0012] FIG. 1B is a rear view showing the external configuration of the portable information terminal of FIG. 1A.

[0013] FIG. 1C is a side view showing the external configuration of the portable information terminal of FIG. 1A.

[0014] FIG. 1D is a partial perspective view showing the external configuration of the portable information terminal of FIG. 1A.

[0015] FIG. 1E is a rear view showing the layout in the rear camera unit of the portable information terminal.

[0016] FIG. 1F is a side view schematically showing the operation of the rear camera unit of the portable information terminal of FIG. 1A.

[0017] FIG. 2A is a block diagram showing the electrical configuration of the portable information terminal equipped with FIG. 1A.

[0018] FIG. 2B is a block diagram showing the electrical configuration of the portable information terminal equipped with FIG. 1A.

[0019] FIG. 2C is a block diagram showing the electrical configuration of the portable information terminal equipped with FIG. 1A.

[0020] FIG. 3A is a diagram showing an example of a user application screen.

[0021] FIG. 3B is a diagram showing an example of an image view window.

[0022] FIG. 3C is a diagram showing an example of an image view window.

[0023] FIG. 3D is a diagram showing an example of the display mode between the user application screen and the image view window.

[0024] FIG. 4A is a diagram schematically showing the alignment by the virtual aimer.

[0025] FIG. 4B is a diagram schematically showing the alignment by the aimer light.

[0026] FIG. 5A is a flowchart showing an example of a multiple code reading mode executed by the service unit.

[0027] FIG. 5B is a flowchart showing an example of the necessity determination of registration executed by the user application according to the multiple code reading mode of FIG. 5A.

[0028] FIG. 5C is a diagram schematically showing the operation executed in the multiple code reading mode of FIG. 5A.

[0029] FIG. 6A is a flowchart showing an example of code reading executed by the portable information terminal.

[0030] FIG. 6B is a flowchart showing an example of a single code reading mode executed in the code reading of FIG. 6A.

[0031] FIG. 6C is a flowchart showing an example of a multiple code reading mode executed in the operation of FIG. 6A.

[0032] FIG. 6D is a flowchart showing an example of data saving process executed in the code reading of FIG. 6A.

[0033] FIG. 6E is a flowchart showing an example of screenshot control.

[0034] FIG. 7A is a flowchart showing an example of code reading condition control.

[0035] FIG. 7B is a diagram schematically showing the operation executed in the code reading condition control of FIG. 7A.

[0036] FIG. 7C is a diagram schematically showing the operation executed in the code reading condition control of FIG. 7A.

[0037] FIG. 7D is a diagram schematically showing the operation executed in the code reading condition control of FIG. 7A.

[0038] FIG. 8A is a rear view showing an example of an attachment.

[0039] FIG. 8B is a rear view showing an example of an attachment.

[0040] FIG. 8C is a rear view showing an example of an attachment.

[0041] FIG. 8D is a rear view showing an example of an attachment.

[0042] FIG. 8E is a rear view showing an example of an attachment.

[0043] FIG. 9 is a block diagram showing the electrical configuration of the portable information terminal that determines the type of attachment.

[0044] FIG. 10 is a flowchart showing a modified example of the method for setting the security mode.

[0045] FIG. 11A is a rear view showing a modified example of the layout in the rear camera unit.

[0046] FIG. 11B is a rear view showing a modified example of the layout in the rear camera unit.

[0047] FIG. 11C is a rear view showing a modified example of the layout in the rear camera unit.

[0048] FIG. 11D is a rear view showing a modified example of the layout in the rear camera unit.

[0049] FIG. 12A is a diagram showing the first modified example of the display mode between the user application screen and the image view window.

[0050] FIG. 12B is a diagram showing the first modified example of the display mode between the user application screen and the image view window.

[0051] FIG. 13A is a diagram showing the second modified example of the display mode between the user application screen and the image view window.

[0052] FIG. 13B is a diagram showing the second modified example of the display mode between the user application screen and the image view window.

[0053] FIG. 14A is a flowchart showing the first modified example of the display control of the image view window.

[0054] FIG. 14B is a flowchart showing the second modified example of the display control of the image view window.

[0055] FIG. 15 is a flowchart showing an example of registration management of read data by the user application.

DETAILED DESCRIPTION

[0056] FIG. 1A is a front view showing the external configuration of a portable information terminal according to the present disclosure, FIG. 1B is a rear view showing the external configuration of the portable information terminal in FIG. 1A, FIG. 1C is a side view showing the external configuration of the portable information terminal in FIG. 1A, FIG. 1D is a partial perspective view showing the external configuration of the portable information terminal in FIG. 1A, FIG. 1E is a rear view showing the layout of the rear camera unit of the portable information

terminal, and FIG. 1F is a side view schematically showing the operation of the rear camera unit of the portable information terminal in FIG. 1A. In this embodiment, the short side direction Xs, long side direction Xl, and thickness direction Xt of the portable information terminal **1**, which are mutually orthogonal, are appropriately indicated.

[0057] The portable information terminal **1** has a rectangular housing **11** with rounded corners when viewed from the front or back in the thickness direction Xt. The housing **11** is short in the short side direction Xs and long in the long side direction Xl; in other words, the size of the housing **11** in the short side direction Xs is shorter compared to the size of the housing **11** in the long side direction Xl. The housing **11** includes a front surface **12** provided on the front side in the thickness direction Xt, a back surface **13** provided on the rear side in the thickness direction Xt, a right side surface **14** provided on the right side when viewed from the front, a left side surface **15** provided on the left side when viewed from the front, a top surface **16** provided on the upper side when viewed from the front, and a bottom surface **17** provided on the lower side when viewed from the front. The front surface **12** and the back surface **13** are parallel or substantially parallel to each other and face each other with a gap in the thickness direction Xt. The right end of the front surface **12** and the right end of the back surface **13** are connected by the right side surface **14** provided in the thickness direction Xt, the left end of the front surface **12** and the left end of the back surface **13** are connected by the left side surface **15** provided in the thickness direction Xt, the upper end of the front surface **12** and the upper end of the back surface **13** are connected by the top surface **16** provided in the thickness direction Xt, and the lower end of the front surface **12** and the lower end of the back surface **13** are connected by the bottom surface **17** provided in the thickness direction Xt.

[0058] The portable information terminal **1** is equipped with a display **20** attached to the housing **11**. The display **20** is a touch panel display provided on the front surface **12** of the housing **11**, and the display **20** displays a screen to the user using the portable information terminal **1** and accepts input in response to touch operations on the display **20**.

[0059] Furthermore, the portable information terminal **1** has volume buttons **21** and **22** and a trigger button **23** arranged on the right side surface **14** of the housing **11**. The right side surface **14** of the housing **11** has a recessed portion **141** that is recessed inward toward the inside of the housing **11**. This recessed portion **141** is extended in the longitudinal direction Xl. The volume buttons **21** and **22** and the trigger button **23** are arranged within the recessed portion **141**, aligned in the longitudinal direction Xl in that order from the top. These volume buttons **21** and **22** and the trigger button **23** are positioned above the center M11 of the longitudinal direction Xl of the housing **11**. When the volume button **21** is operated by the user, the volume of the speaker of the portable information terminal **1** increases, and when the volume button **22** is operated by the user, the volume of the speaker of the portable information terminal **1** decreases. The trigger button **23** provides a trigger to the portable information terminal **1** to start reading a code when the trigger button **23** is operated by the user. The trigger button **23** has a base portion **231** that is arranged within the recessed portion **141** and extends in the longitudinal direction Xl, and a protruding portion **232** that protrudes outward from the recessed portion **141** from the base portion **231**. In the longitudinal direction Xl, the protruding portion **232** is shorter than the base portion **231** and is provided at the center of the base portion **231**. This base portion **231** protrudes outward (to the right) from the right side surface **14** (FIG. 1A, FIG. 1B).

[0060] Furthermore, the portable information terminal **1** is equipped with a camera module **3** attached to the housing **11**. The camera module **3** has a front camera **31** provided on the front surface **12** of the housing **11** and a rear camera unit **32** provided on the rear surface **13** of the housing **11**.

[0061] The front camera **31** has a camera field (front view) that expands forward from the front surface **12** of the housing **11**, and captures the field using a rolling shutter method. In FIG. 1A, the objective lens of the front camera **31** appearing on the front surface **12** of the housing **11** is shown.

The front camera **31** is an autofocus camera that automatically adjusts the focus position. The autofocus is executed by adjusting the focus position based on the brightness histogram of the image captured by the front camera **31**. The front camera **31** may also be a fixed focus camera. [0062] The rear camera unit **32** is provided protruding from the rear surface **13** of the housing **11** to the rear side in the thickness direction X_t , and has a rear camera **33**, a wide-angle illumination unit **34**, a narrow-angle illumination unit **35**, an aimer light illumination unit **36**, and a distance measuring sensor **37**. The rear camera **33** has a camera field V_{33} (rear view) that expands to the rear side from the rear surface **13** of the housing **11**, and captures the field using a rolling shutter method. The rear camera **33** is, for example, a color camera capable of generating color images. In FIG. **1B**, the objective lens of the rear camera unit **32** appearing on the rear surface **13** of the housing **11** is shown. The rear camera unit **32** is an autofocus camera that automatically adjusts the position of the focus. When using the rear camera unit **32** for photography, autofocus is executed based on the brightness histogram, similar to the front camera **31**. Also, when using the rear camera unit **32** for code reading, autofocus is executed based on the detection result of the distance measuring sensor **37**, as described later. The wide-angle illumination unit **34** emits wide-angle illumination light L_{34} , which is illumination light with a relatively wide light distribution angle, to the rear side from the rear surface **13** of the housing **11**. The narrow-angle illumination unit **35** emits narrow-angle illumination light L_{35} , which is illumination light with a narrower light distribution angle than the wide-angle illumination light L_{34} , to the rear side from the rear surface **13** of the housing **11**. The aimer light illumination unit **36** emits aimer light L_{36} , which is visible light, to the rear side from the rear surface **13** of the housing **11**. The distance measuring sensor **37** is, for example, a TOF (Time of Flight) sensor, and emits distance measuring light L_{37} to the rear side from the rear surface **13** of the housing **11** and measures distance to measurement target based on time from emitting the distance measuring light L_{37} until detecting the distance measuring light L_{37} reflected by measurement target.

[0063] As shown in FIG. **1E**, the wide-angle illumination unit **34**, distance measuring sensor **37**, rear camera **33**, and aimer light illumination unit **36** are arranged in this order in the short direction X_s . In other words, in the short direction X_s , the aimer light illumination unit **36** is positioned on one side of the rear camera **33**, while the distance measuring sensor **37** and wide-angle illumination unit **34** are positioned on the other side of the rear camera **33**. The rear camera **33** is located between the distance measuring sensor **37** and the aimer light illumination unit **36**, and the distance measuring sensor **37** is located between the rear camera **33** and the wide-angle illumination unit **34**. The narrow-angle illumination unit **35** is positioned below the rear camera **33** in the long direction X_l . The optical axes of the rear camera **33**, wide-angle illumination unit **34**, narrow-angle illumination unit **35**, aimer light illumination unit **36**, and distance measuring sensor **37** are extended in the thickness direction X_t and are parallel or substantially parallel to each other. The optical axis of the rear camera **33** is separated from the optical axis of the wide-angle illumination unit **34** by a distance K_{34} , the optical axis of the rear camera **33** is separated from the optical axis of the narrow-angle illumination unit **35** by a distance K_{35} , the optical axis of the rear camera **33** is separated from the optical axis of the aimer light illumination unit **36** by a distance K_{36} , and the optical axis of the rear camera **33** is separated from the optical axis of the distance measuring sensor **37** by a distance K_{37} . Here, the optical axis of the distance measuring sensor **37** is a virtual axis that extends parallel or substantially parallel to each optical axis, positioned midway between the optical axis of the optical system that emits the distance measuring light L_{37} and the optical axis of the system that focuses the distance measuring light L_{37} on to the optical sensor. Furthermore, the distance K_{34} between the rear camera **33** and the wide-angle illumination unit **34** is longer than the distances K_{35} , K_{36} , and K_{37} between the rear camera **33** and the narrow angle illumination unit **35**, the aimer light illumination unit **36**, and the distance measuring sensor **37**, respectively. [0064] As shown in FIG. **1F**, the rear camera **33** has a camera field of view V_{33} that extends to the rear side in the thickness direction X_t . In contrast, the wide-angle illumination unit **34** irradiates

wide-angle illumination light L34 within the camera field of view V33, and the narrow-angle illumination unit 35 irradiates narrow-angle illumination light L35 within the camera field of view V33. The light distribution angle of the narrow-angle illumination light L35 is smaller than that of the wide-angle illumination light L34, and the irradiation range of the narrow-angle illumination light L35 is included in the irradiation range of the wide-angle illumination light L34. The aimer light illumination unit 36 irradiates aimer light L36 within the camera field of view V33, and the distance measuring sensor 37 irradiates distance measuring light L37 within the camera field of view V33. The aimer light L36 and the distance measuring light L37 are irradiated at approximately the center of the camera field of view V33, and the irradiation ranges of the aimer light L36 and the distance measuring light L37 are included in the irradiation range of the wide-angle illumination light L34 and the irradiation range of the narrow-angle illumination light L35.

[0065] FIG. 2A is a block diagram showing the electrical configuration of the portable information terminal shown in FIG. 1A, mainly illustrating the configuration for controlling the camera module. The portable information terminal 1 includes a control unit 5 that controls the camera module 3, a volatile memory unit 61, and a non-volatile memory unit 62. The control unit 5 is, for example, a processor that controls the camera module 3 and manages data storage to the volatile memory unit 61 and the non-volatile memory unit 62. The volatile memory unit 61 is, for example, RAM (Random Access Memory) that temporarily stores data. The non-volatile memory unit 62 is, for example, an ROM (Read-Only Memory), Flash Memory such as NAND flash memory, SSD (Solid State Drive) or HDD (Hard Disk Drive) that permanently stores data. Furthermore, the portable information terminal 1 is equipped with a photo shooting processing unit 71 that executes photo shooting by the control unit 5 using the camera module 3, and a code reading processing unit 72 that executes code reading by the control unit 5 using the camera module 3. The photo shooting processing unit 71 and the code reading processing unit 72 are processes generated by the processor mounted in the portable information terminal 1 executing a predetermined program.

[0066] The photo shooting processing unit 71 saves the setting values (photo shooting setting values) set in the camera module 3 for taking pictures using the front camera 31 or the rear camera 33 in the non-volatile memory unit 62. The photo shooting setting values include, for example, the exposure time of the front camera 31 or the rear camera 33, the brightness of the wide-angle illumination light L34 emitted from the wide-angle illumination unit 34 to the camera field of view V33, and the focus position of the front camera 31 or the rear camera 33. The code reading processing unit 72 saves the setting values (code reading setting values) set in the camera module 3 for reading codes using the rear camera 33 in the non-volatile memory unit 62. The code reading setting values include, for example, the exposure time of the rear camera 33, the brightness of the narrow-angle illumination light L35 emitted from the narrow-angle illumination unit 35 to the camera field of view V33, the focus position of the rear camera 33, and the size of the code that is the reading target.

[0067] The control unit 5 has a camera control public API 51, a camera control private API 52, an API execution switching control unit 53, and a camera control unit 54, where the camera control unit 54 has a control unit 55 for public API and a control unit 56 for private API. These units of the control unit 5 are processes generated by a processor mounted in the portable information terminal 1 executing a predetermined program in the non-volatile memory unit 62.

[0068] When the Camera Control Public API 51 receives a photo capture instruction that instructs the execution of photo shooting from the Photo Shooting Processing Unit 71, the API Execution Switching Control Unit 53 manages the execution of photo shooting based on the determination result of whether the security mode is set or not. The setting of the security mode is executed by the administrator's operation on the Security Setting Unit 74, which will be described later. If the security mode is not set (in other words, if the normal mode is set), the API Execution Switching Control Unit 53 permits photo shooting and sends the photo capture instruction to the Control Unit for Public API 55. Therefore, the Control Unit for Public API 55 reads the photo capture setting

value from the Non-volatile Memory Unit **62** and operates the Camera Module **3** based on this photo capture setting value. In other words, the Control Unit for Public API **55** acquires the photo image Ip (captured image) captured by the front camera **31** or the rear camera **33**, and stores the photo image Ip in the Non-volatile Memory Unit **62**. On the other hand, if the security mode is set, the API Execution Switching Control Unit **53** prohibits photo shooting and does not send the photo capture instruction to the Control Unit for Public API **55**. Therefore, the Control Unit for Public API **55** does not execute photo shooting using the Camera Module **3**.

[0069] When the camera control private API **52** receives a code reading instruction that instructs the execution of code reading from the code reading processing unit **72**, the API execution switching control unit **53** sends the code reading instruction to the private API control unit **56**. Therefore, the private API control unit **56** reads the code reading setting value from the non-volatile memory unit **62** and operates the camera module **3** based on this code reading setting value. In other words, the private API control unit **56** acquires the code image Ic (captured image) captured by the rear camera **33** in the camera field of view **V33** and stores it in the volatile memory unit **61**. As will be described later, a decoding process is executed on the code image Ic stored in the volatile memory unit **61**. Furthermore, if the administrator has set the non-volatile storage of the code image Ic, the private API control unit **56** saves the code image Ic in the non-volatile memory unit **62**, while if the administrator has not set the non-volatile storage of the code image Ic, the private API control unit **56** does not save the code image Ic in the non-volatile memory unit **62**.

[0070] In addition, the portable information terminal **1** is equipped with an SoC (System on Chip) **73**. This SoC **73** includes an ISP (Image Signal Processor) **731**, a DSP (Digital Signal Processor) **732**, a GPU (Graphic Processing Unit) **733**, and a CPU (Central Processing Unit) **734**. The ISP **731** receives a photo image Ip or a code image Ic from the front camera **31** or the rear camera **33** of the camera module **3** and performs specific corrections to these images that are unique to the front camera **31** or the rear camera **33**. The specific corrections for the camera correspond to the correction of errors caused by the characteristics of the camera's optical system and image sensor. Thus, the SoC **73** executes appropriate signal processing on the photo image Ip or the code image Ic corrected by the ISP **731** using the DSP **732**, GPU **733**, or CPU **734**. The control unit for the public API **55** and the control unit for the private API **56** store the photo image Ip or the code image Ic, on which signal processing has been performed by the SoC **73**, in the non-volatile memory unit **62** or the volatile memory unit **61**.

[0071] Furthermore, the portable information terminal **1** is equipped with a security setting unit **74**, a publication prohibition unit **75**, a screenshot prohibition unit **76**, and a screenshot control unit **77**. The security setting unit **74** displays the administrator setting screen on the display **20** and accepts the administrator's setting operation for the display **20**. Specifically, when the administrator performs an operation to instruct the setting of the security mode on the administrator setting screen, the security setting unit **74** sets the security mode, and when the administrator performs an operation to instruct the release of the security mode on the administrator setting screen, the security setting unit **74** releases the security mode. The security setting unit **74** displays the administrator setting screen on the display **20** when the security setting unit **74** confirms that a predetermined password has been entered on the display **20**. The publication prohibition unit **75** prohibits the publication of the photo image Ip or the code image Ic captured by the front camera **31** or the rear camera **33** when the security mode is set.

[0072] The screenshot prohibition unit **76** prohibits screenshots of the display **20** when the security mode is set. The screenshot control unit **77** does not execute the screenshot if the screenshot is prohibited by the screenshot prohibition unit **76**, ignoring the execution instruction for the screenshot. Instead of prohibiting the screenshot, the screenshot control unit **77** may save an image in the non-volatile memory unit **62** that is either disabled (replacing all pixel values with a constant pixel value) or masked (replacing it with an image composed of a mask with a constant pixel value).

[0073] Furthermore, the portable information terminal **1** has a location information calculation unit **78**. The location information calculation unit **78** calculates the location information of the portable information terminal **1** using GNSS (Global Navigation Satellite System). The location information of the portable information terminal **1** calculated by the location information calculation unit **78** is stored in the volatile memory unit **61**.

[0074] FIG. 2B is a block diagram showing the electrical configuration of the portable information terminal shown in FIG. 1A, mainly illustrating the configuration that controls communication with the outside of the portable information terminal **1**. The portable information terminal **1** includes a communication control unit **81**, an internet communication unit **82**, a telephone communication unit **83**, and a call recording unit **84**. The internet communication unit **82** communicates with external servers via the internet, the telephone communication unit **83** conducts calls with external telephone terminals through communication lines (including telephone lines, LAN lines, and internet lines), and the call recording unit **84** stores the voice data of calls made by the telephone communication unit **83** in the non-volatile memory unit **62**. Furthermore, the operations of the internet communication unit **82**, the telephone communication unit **83**, and the call recording unit **84** are controlled based on the determination result by the communication control unit **81** regarding whether the security mode is set or not.

[0075] That is, the communication control unit **81** permits the internet connection by the internet communication unit **82** when it is determined that the security mode is not set. The communication control unit **81** prohibits the internet connection by the internet communication unit **82** when it is determined that the security mode is set.

[0076] In addition, the communication control unit **81** has a registered telephone number list indicating registered telephone numbers. If the communication control unit **81** determines that the security mode is not set, it allows calls from both the telephone terminal of the telephone number included in the registered telephone number list and the telephone terminal of the telephone number not included in the registered telephone number list via the telephone communication unit **83**. If the communication control unit **81** determines that the security mode is set, it allows calls between the telephone terminal of the telephone number included in the registered telephone number list and the telephone communication unit **83**, while prohibiting calls between the telephone terminal of the telephone number not included in the registered telephone number list and the telephone communication unit **83**.

[0077] The communication control unit **81** permits the storage of voice data in the non-volatile memory unit **62** by the call recording unit **84** when it is determined that the security mode is not set. The communication control unit **81** prohibits the storage of voice data in the non-volatile memory unit **62** by the call recording unit **84** when it is determined that the security mode is set.

[0078] FIG. 2C is a block diagram showing the electrical configuration of the portable information terminal shown in FIG. 1A, primarily illustrating the configuration for controlling code reading by the portable information terminal **1**. The portable information terminal **1** includes a decoder **86** (image processing unit), a user application **87**, and a service unit **88**. Each of the decoder **86**, user application **87**, and service unit **88** is a process generated by a processor mounted in the portable information terminal **1** executing a predetermined program. The service unit **88** functions as a backend that coordinates the operations of the decoder **86**, user application **87**, code reading processing unit **72**, trigger button **23**, and display **20**.

[0079] The service unit **88** confirms that the trigger button **23** has been pressed by the user, sends a code reading instruction to the code reading processing unit **72**, and the code reading processing unit **72** outputs the code reading instruction received from the service unit **88** to the camera control private API **52**. In response to receiving the code reading instruction from the camera control private API **52**, the code image **Ic** is obtained by the camera module **3** according to the aforementioned procedure. That is, the control unit for the private API **56** sets the code reading setting value read from the non-volatile memory unit **62** to the camera module **3**. Then, the control

unit for the private API **56** causes the narrow-angle illumination light **L35** to be emitted from the narrow-angle illumination unit **35** while capturing the camera field of view **V33** with the rear camera **33**. At this time, the control unit for the private API **56** adjusts the focus position of the rear camera **33** to match the position of the code on the workpiece based on the result of detecting the distance to the code (specifically, the workpiece with the code) measured by the distance measuring sensor **37**. Thus, when the focus of the rear camera **33** is in a state matched to workpiece, the irradiation positions of the aimer light **L36** and the distance measuring light **L37** on the workpiece **W** are included in depth of field of the rear camera **33**. Then, the control unit for private API **56** executes imaging with the rear camera **33**. As a result, the entire image of the camera field of view **V33** containing code is obtained as the code image **Ic**, and the code image **Ic** is stored in the volatile memory unit **61**. At this time, the control unit for private API **56** repeatedly acquires code image **Ic** at a predetermined frame cycle and overwrites it in volatile memory unit **61**. Therefore, only latest code image **Ic** is stored in the volatile memory unit **61**.

[0080] Then, the service unit **88** confirms that the trigger button **23** has been pressed by the user and, after a predetermined waiting time (for example, 1 second) has elapsed, requests the decoder **86** to execute the decoding process at that timing. As a modified example, the service unit **88** may request the decoder **86** to execute the decoding process at the timing when the trigger button **23** is pressed by the user (that is, without waiting time). The decoder **86**, upon receiving the request from the service unit **88**, reads the code image **Ic** stored in the volatile memory unit **61** and executes the decoding process on the code image **Ic**. As a result, the reading data **Dc** indicated by the code image **Ic** is read by the decoder **86**. This reading data **Dc** is, for example, text data.

[0081] The decoder **86** sends the reading data **Dc** to the service unit **88**, and the service unit **88** sends the reading data **Dc** received from the service unit **88** to the user application **87**. The user application **87** inputs the reading data **Dc** received from the decoder **86** into a predetermined input item. At this time, the user application **87** displays a user application screen **91** (FIG. 3A) on the display **20** to assist the user in inputting the reading data **Dc**.

[0082] FIG. 3A is a diagram showing an example of a user application screen. The user application screen **91** in FIG. 3A is displayed on the display **20** in conjunction with the execution of the user application **87**. The user application screen **91** has multiple input fields **911** that each indicate input items **A1**, **A2**, . . . , and multiple data display fields **912** that each indicate reading data **Dc** input into the multiple input items **A1**, **A2**, In the example of FIG. 3A, reading data **Dc1** and **Dc2** have been input for input items **A1** and **A2**, respectively, while reading data **Dc** has not been input for input items **A3**, **A4**, In addition, the user application **87** is focusing on the input item **A3** selected by the user's operation on the display **20**. When the user presses the trigger button **23** in this state, reading data **Dc3** is obtained by the service unit **88** according to the aforementioned procedure and sent to the user application **87**. Then, the user application **87** inputs the reading data **Dc3** received from the service unit **88** into the focused input item **A3**. As a result, the reading data **Dc3** is displayed in the data display field **912** corresponding to the input item **A3** on the user application screen **91**.

[0083] Furthermore, the service unit **88** displays an image view window **93** (FIG. 3B, FIG. 3C) on the display **20** to assist the user in reading the code. FIGS. 3B and 3C show an example of the image view window. The image view window **93** has a rectangular shape and displays the code image **Ic** stored in the volatile memory unit **61**. The image view window **93** is an image view window (also called a live view window) that updates the image by capturing images periodically. Not limited to this, the image view window **93** may also update the screen with the captured image when the decoder **86** successfully reads it. This image view window **93** is displayed on the display **20** by the service unit **88** in conjunction with the activation of the rear camera **33** of the camera module **3**. Therefore, for example, in response to the pressing of the trigger button **23**, the service unit **88** displays the image view window **93** on the display **20**. In other words, when the service unit **88** confirms the pressing of the trigger button **23**, it sends a code reading instruction to the code

reading processing unit **72**. When this code reading instruction is received by the code reading processing unit **72**, the rear camera **33** is activated for code reading according to the aforementioned procedure, and the display of the image view window **93** on the display **20** begins. [0084] As described above, when the trigger button **23** is pressed by the user, the code image Ic is acquired at a predetermined frame cycle and overwritten in the volatile memory **61**. In contrast, the service unit **88** reads the code image Ic stored in the volatile memory **61** at the frame cycle, acquires the image of the central portion of the code image Ic as the image view image Il, and displays it in the image view window **93** currently displayed on the display **20**. In the examples of FIG. 3B and FIG. 3C, the code C included in the camera field of view V**33** is projected in the image view image Il.

[0085] Furthermore, the service unit **88** displays the virtual aimer **94** at the center of the image view window **93**. Therefore, the user holding the portable information terminal **1** can align the virtual aimer **94** displayed in the image view image Il with the code C. As a result, the virtual aimer **94**, which is offset from the code C as shown in FIG. 3B, can be matched with the code C as shown in FIG. 3C. This allows the code C to be positioned at the center of the camera field of view V**33**. Then, when the decoder **86** completes the decoding process for the code image Ic, the service unit **88** ends the display of the image view image Il on the display **20**. The display of the image view image Il starts simultaneously with the confirmation of the pressing of the trigger button **23**, while the decoding process of the decoder **86** starts after the waiting time has elapsed from the confirmation of the pressing of the trigger button **23**. Therefore, the image view image Il is continuously displayed on the display **20** for at least the waiting time.

[0086] Furthermore, the service unit **88** displays the image view window **93** on the display **20** together with the user application screen **91** displayed by the user application **87** (FIG. 3D). Here, FIG. 3D is a diagram showing an example of the display mode of the user application screen and the image view window. As shown in FIG. 3D, the service unit **88** displays the image view window **93** over the user application screen **91**. In other words, the image view window **93** is displayed in front of the user application screen **91**. At this time, the service unit **88** displays the image view window **93** in a semi-transparent manner. Therefore, the overlapping portion of the user application screen **91** with the image view window **93** is displayed in a semi-transparent manner. In other words, the overlapping portion is displayed on the display **20** so that it can be visually recognized by the user. In the example of FIG. 3D, the input item A**3**, which is focused by the user, is included in the overlapping portion, but it is visually recognizable by the user because it is displayed in a semi-transparent manner through the image view window **93**.

[0087] At this time, the service unit **88** adjusts the transparency rate, which is the degree to which the image view window **93** allows the user application screen **91** to be seen through. Specifically, the service unit **88** calculates a representative value that represents the saturation of the image in the image view window **93**. For example, the maximum value, average value, or median value of the saturation in the image view window **93** is calculated as the representative value. Then, the service unit **88** sets a larger transparency rate as the saturation (representative value) of the image displayed in the image view window **93** increases.

[0088] In addition, along with the display of the virtual aimer **94** on the image view image Il, the irradiation of the aimer light L**36** by the aimer light illumination unit **36** is also executed. That is, when the private API control unit **56** receives a code reading instruction from the code reading processing unit **72**, it causes the aimer light illumination unit **36** to irradiate the aimer light L**36**. Therefore, the user can perform alignment between the camera field of view V**33** and the code C by the virtual aimer **94** as described above, and can also perform alignment between the camera field of view V**33** and the code C by the aimer light L**36** (in other words, the real aimer) irradiated on the workpiece with the code C attached (FIG. 4A, FIG. 4B).

[0089] FIG. 4A is a schematic diagram showing the alignment state using a virtual aimer, and FIG. 4B is a schematic diagram showing the alignment state using aimer light. In both examples of FIG.

4A and FIG. 4B, the aimer light L36 emitted from the aimer light illumination unit 36 is irradiated onto the workpiece W with the code C attached. However, in the example of FIG. 4A, the user's line of sight E is hidden by the portable information terminal 1 and does not reach the aimer light L36 on the workpiece W. Therefore, the user cannot visually recognize the aimer light L36 and performs alignment based on the virtual aimer 94 displayed on the display 20. On the other hand, in the example of FIG. 4B, the user's line of sight E reaches the aimer light L36 on the workpiece W without being hidden by the portable information terminal 1. Therefore, even if the user cannot visually recognize the display 20, the user can visually recognize the aimer light L36 and perform alignment based on the aimer light L36 irradiated onto the workpiece W.

[0090] Furthermore, the portable information terminal 1 is capable of executing a single code reading mode that reads a single code C as illustrated in FIGS. 3B and 3C, and a multiple code reading mode that reads multiple codes C simultaneously. Next, the multiple code reading mode will be described.

[0091] FIG. 5A is a flowchart showing an example of a multiple code reading mode executed by the service unit, and FIG. 5B is a flowchart showing an example of registration necessity determination executed by the user application according to the multiple code reading mode of FIG. 5A, and FIG. 5C is a diagram schematically showing the operation executed in the multiple code reading mode of FIG. 5A. In FIG. 5C, the operation of reading six codes C1 to C6 attached to a workpiece W, which is a form, is exemplified. In this example, the reading data Dc, which is the result of reading the code C, is registered in the data management table 95 held by the user application 87.

[0092] This data management table 95 shows the data registration conditions 951 that the reading data Dc must satisfy and the reading results 952 that serve as the registration destination for the reading data Dc, for each of the six reading targets Tx1 to Tx6. The data registration conditions 951 are composed of a combination of code type, character string, and position. The code type indicates the code type conditions required for the type of code C corresponding to the reading data Dc, the character string indicates the character string conditions required for the character strings included in the reading data Dc, and the position indicates the position conditions required for the position of the code C corresponding to the reading data Dc in the workpiece W. In this example, the position condition is N/C (No Care) and is therefore invalid. It should be noted that the conditions included in the data registration conditions 951 are not limited to this example. The data registration conditions 951 may also include numerical conditions required for the numerical values included in the reading data Dc.

[0093] In step S101, the service unit 88 has the decoder 86 perform decoding processing on the code image Ic obtained by imaging the code C included in the image view window 93, thereby acquiring the reading data Dc of the code C. In the example shown in the column "S101 (1st time)" of FIG. 5C, out of the six codes C1 to C6, four codes C1 to C4 are included in the image view window 93. Therefore, the code image Ic including the four codes C1 to C4 is obtained and stored in the volatile memory 61, and the decoder 86 performs decoding processing on the images of each of the codes C1 to C4 included in the code image Ic. As a result, the service unit 88 acquires four reading data Dc1 to Dc4 corresponding to the four codes C1 to C4, respectively.

[0094] In step S102, the service unit 88 sends the four pieces of reading data Dc1 to Dc4 obtained in step S101 to the user application 87. In response, the user application 87 executes the necessity determination for registration shown in FIG. 5B. It should be noted that the user application 87 performs the necessity determination for registration for each of the four pieces of reading data Dc1 to Dc4, but here, the necessity determination for registration for one piece of reading data Dc is explained.

[0095] User application 87 receives reading data Dc from service unit 88 (step S201), and collates reading data Dc with data management table 95 to determine the necessity of registering reading data Dc into data management table 95 (steps S202 to S205). In step S202, user application 87

checks whether there exists any data registration condition **951** among the data registration conditions **951** of reading targets Tx1 to Tx6 that reading data Dc satisfies. If no applicable data registration condition **951** exists (in step S202, if “NO”), user application **87** determines that reading data Dc is not a reading target and thus registration is unnecessary, and sends that determination result to service unit **88** (step S205).

[0096] If the corresponding data registration condition **951** exists (in the case of “YES” in step S202), the user application **87** determines whether the read data Dc is already registered in the reading result **952** corresponding to the relevant data registration condition **951** (step S203). If it is already registered (in the case of “YES” in step S203), the user application **87** determines that the read data Dc is already registered and therefore unnecessary for registration, and sends that determination result to the service unit **88** (step S205). If it is not registered (in the case of “NO” in step S203), the user application **87** determines that the read data Dc is necessary for registration and sends that determination result to the service unit **88** (step S204).

[0097] In the example shown in the column “S102 (1st time)” of FIG. 5C, each of the reading data Dc1 to Dc4 is determined to satisfy the data registration conditions **951** of the reading targets Tx1 to Tx4 (“YES” in step S202). Furthermore, since the reading results **952** corresponding to each of the reading targets Tx1 to Tx4 are empty and the reading data Dc1 to Dc4 are unregistered, the reading data Dc1 to Dc4 is determined to be necessary for registration (steps S203, S204).

[0098] As shown in FIG. 5A, when the service unit **88** receives the necessity determination result from the user application **87** (step S103), it attaches a mark corresponding to the determination result to codes C1 to C4 in the image view window **93** (step S104). In this example, since codes C1 to C4 (in other words, the corresponding reading data Dc1 to Dc4) are determined to require registration, the service unit **88** displays a mark Ma indicating that registration is necessary (for example, a black check mark) for codes C1 to C4 (in the column of “S104 (1st time)” in FIG. 5C).

[0099] The service unit **88** confirms the user's flick operation (registration instruction operation) on the display **20** where the image view window **93** is displayed (step S105), and sends a registration instruction to the user application **87** to register the reading data Dc1 to Dc4, which is determined to be necessary for registration, into the reading result **952** of the data management table **95** (step S106). When the user application **87** receives the registration instruction from the service unit **88**, it registers the reading data Dc1 to Dc4 indicated by the registration instruction into the reading results **952** of the corresponding reading targets Tx1 to Tx4 (the column of “registration (first time)” in FIG. 5C).

[0100] In the example shown in the column “S101 (second time)” of FIG. 5C, out of the six codes C1 to C6, four codes C3 to C6 are included in the image view window **93**, and the service unit **88** acquires four reading data Dc3 to Dc6 corresponding to the four codes C3 to C6.

[0101] In step S102, the service unit **88** sends the four pieces of reading data Dc3 to Dc6 obtained in step S101 to the user application **87**. In response, the user application **87** executes the necessity determination for registration shown in FIG. 5B. In the example shown in the column “S102 (2nd time)” of FIG. 5C, each of the reading data Dc3 to Dc6 is determined to satisfy the data registration conditions **951** of the reading targets Tx3 to Tx6 (“YES” in step S202).

[0102] However, since the reading data Dc3 and Dc4 corresponding to the reading targets Tx3 and Tx4 are already registered in the reading result **952**, the reading data Dc3 and Dc4 are determined to be unnecessary for registration (steps S203, S205). Therefore, the determination result that the reading data Dc3 and Dc4 are unnecessary for registration because they are already registered is sent from the user application **87** to the service unit **88**. On the other hand, since the reading results **952** corresponding to the reading targets Tx5 and Tx6 are empty and the reading data Dc5 and Dc6 are unregistered, the reading data Dc5 and Dc6 are determined to be necessary for registration (steps S203, S204). Therefore, the determination result that the reading data Dc5 and Dc6 are necessary for registration is sent from the user application **87** to the service unit **88**.

[0103] As shown in FIG. 5A, the service unit **88** receives the necessity determination result from

the user application **87** (step **S103**) and attaches a mark corresponding to the determination result to codes **C3** to **C6** in the image view window **93**. In this example, codes **C3** and **C4** (in other words, the corresponding reading data **Dc3** and **Dc4**) are determined to be unnecessary for registration because they are already registered. Therefore, the service unit **88** displays a mark **Mb** (for example, a white check mark) indicating that they are registered for codes **C3** and **C4** (in the column of “**S104** (second time)” in FIG. **5C**). Additionally, codes **C5** and **C6** (in other words, the corresponding reading data **Dc5** and **Dc6**) are determined to require registration. Therefore, the service unit **88** displays a mark **Ma** indicating that registration is necessary for codes **C5** and **C6** (in the column of “**S104** (second time)” in FIG. **5C**).

[0104] In the example of FIG. **5C**, a mark **Ma** indicating the necessity of registration and a mark **Mb** indicating that registration is unnecessary because it is already registered are shown, but the types of marks are not limited to these. In other words, when the reading data **Dc** is determined to be unnecessary for registration because it is not a reading target, a mark of a different form from marks **Ma** and **Mb** may be displayed as a mark indicating this determination result.

[0105] The service unit **88** confirms the user's flick operation (registration instruction operation) on the display **20** where the image view window **93** is displayed (step **S105**), and sends a registration instruction to the user application **87** to register the reading data **Dc5** and **Dc6**, which are determined to require registration, into the reading result **952** of the data management table **95** (step **S106**). When the user application **87** receives the registration instruction from the service unit **88**, it registers the reading data **Dc5** and **Dc6** indicated by the registration instruction into the reading results **952** of the corresponding reading targets **Tx5** and **Tx6** (the “registration (second time)” column in FIG. **5C**).

[0106] FIG. **6A** is a flowchart showing an example of code reading executed by a portable information terminal, and FIG. **6B** is a flowchart showing an example of a single code reading mode executed in the code reading of FIG. **6A**, and FIG. **6C** is a flowchart showing an example of a multiple code reading mode executed in the operation of FIG. **6A**, and FIG. **6D** is a flowchart showing an example of data saving process executed in the code reading of FIG. **6A**.

[0107] In step **S301** of FIG. **6A**, the main body of the portable information terminal **1** is activated, and the display **20** is started. In step **S302**, the camera control unit **54** reads the code reading setting value from the non-volatile memory **62** and sets it to the camera module **3**. In addition, the camera control unit **54** also performs initialization of the ISP **731** and the like in step **S302**. Step **S303** is where the camera control unit **54** puts the camera module **3** to sleep.

[0108] Then, when the camera control private API **52** receives a code reading instruction from the code reading processing unit **72** (“YES” in step **S304**), the control unit for private API **56** checks whether the single code reading mode is set (step **S305**). In other words, the user can set which mode to execute, either the single code reading mode or the multiple code reading mode, to the control unit for private API **56** by operating the display **20**. If the single code reading mode is set (“YES” in step **S305**), the single code reading mode shown in FIG. **6B** is executed (step **S306**). On the other hand, if the multiple code reading mode is set (“NO” in step **S305**), the multiple code reading mode shown in FIG. **6C** is executed (step **S307**).

[0109] In step **S401** of FIG. **6B**, the non-public API control unit **56** starts the single code reading mode. When the service unit **88** confirms the pressing of the trigger button **23** (step **S402**), the non-public API control unit **56** activates the rear camera unit **32** of the camera module **3** (step **S403**). As mentioned above, the image view window **93** is displayed on the display **20** accompanying the activation of the rear camera unit **32**.

[0110] Then, the control unit for private API **56** generates a code image **Ic** by the rear camera unit **32** (step **S404**), and temporarily stores this code image **Ic** in the volatile memory **61** (step **S405**). In step **S406**, the decoder **86** performs decoding processing on the single code **C** included in the code image **Ic**, and the reading data **Dc** is acquired. As mentioned above, the decoding processing is executed at a timing after a predetermined waiting time (for example, 1 second) has elapsed from

the confirmation of the trigger button **23** being pressed, or at the timing of the confirmation of the trigger button **23** being pressed. In step **S407**, the service unit **88** sends the reading data **Dc** acquired from the decoder **86** to the user application **87**. Then, this reading data **Dc** is registered in the data management table **95** of the user application **87**.

[0111] In step **S501** of FIG. **6C**, the private API control unit **56** starts the multiple code reading mode. In step **S502**, the private API control unit **56** activates the rear camera unit **32** of the camera module **3**. As mentioned above, with the activation of the rear camera unit **32**, the image view window **93** is displayed on the display **20**, and the service unit **88** displays the code image **Ic** captured by the rear camera unit **32** in the image view window **93** (step **S503**). At this time, the service unit **88** increases the area of the image view window **93** displayed in the multiple code reading mode compared to the area of the image view window **93** displayed in the single code reading mode. When including multiple codes within the field of view, if the area of the image view window **93** is constant, the size of each code will become smaller, but this configuration can suppress the reduction in size of the codes within the field of view.

[0112] The service unit **88** confirms the pressing of the trigger button **23** (step **S504**), and instructs the decoder **86** to perform the decoding process for the code image **Ic** stored in the volatile memory unit **61** (step **S505**). Since the code image **Ic** contains images of multiple codes **C**, the decoder **86** executes the decoding process for each of the multiple codes **C** to obtain multiple reading data **Dc**. The service unit **88** sends the multiple reading data **Dc** obtained from the decoder **86** to the user application **87** (step **S506**), and through the aforementioned flow using FIG. **5A** and FIG. **5B**, the reading data **Dc** is registered in the data management table **95** of the user application **87**.

[0113] As shown in FIG. **6A**, when the single code reading mode (step **S306**) or the multiple code reading mode (step **S307**) is completed, the non-public API control unit **56** puts the camera module **3** in standby (step **S308**). Then, in step **S309**, the non-public API control unit **56** executes the data saving process of FIG. **6D** (step **S309**).

[0114] In step **S601** of FIG. **6D**, the control unit **56** for the private API determines whether the storage of the code image **Ic** to the non-volatile memory **62** is set or not. In other words, the user can set whether to save the code image **Ic** to the non-volatile memory **62** by operating the display **20**. If the storage to the non-volatile memory **62** is not set (in the case of “NO” in step **S601**), the control unit **56** for the private API deletes the code image **Ic** temporarily stored in the volatile memory **61** from the volatile memory **61** (step **S604**).

[0115] On the other hand, when saving to the non-volatile memory **62** is set (in the case of “YES” in step **S601**), the control unit for private API **56** determines whether the security mode is set (step **S602**). Specifically, the control unit for private API **56** determines based on the result of receiving a mode selection signal indicating which of the security mode and normal mode is selected from the security setting unit **74**. When the security mode is set (in the case of “YES” in step **S602**), the control unit for private API **56** deletes the code image **Ic** from the volatile memory **61** (step **S604**). On the other hand, when the security mode is not set (in the case of “NO” in step **S602**), in other words, when the normal mode is set, the control unit for private API **56** saves the code image **Ic** temporarily stored in the volatile memory **61** to the non-volatile memory **62** (step **S603**), and then deletes the code image **Ic** from the volatile memory **61** (step **S604**).

[0116] In addition, the execution of the screenshot is appropriately restricted for the display **20** that displays the image view window **93**. That is, when the image view window **93** is displayed on the display **20** in the multiple code reading mode, the screenshot control unit **77** accepts the execution instruction of the screenshot and executes the flowchart in FIG. **6E**. Here, FIG. **6E** is a flowchart showing an example of screenshot control.

[0117] The screenshot control unit **77** accepts the execution instruction of the screenshot (if “YES” in step **S701**), and determines whether the security mode is set (step **S702**). Then, if the security mode is not set (if “NO” in step **S702**), the screenshot control unit **77** saves the screenshot image of the display **20** displaying the image view window **93** in the non-volatile memory **62** (step **S703**).

[0118] On the other hand, when the security mode is set (in the case of “YES” in step S702), as mentioned above, the screenshot prohibition unit **76** prohibits the screenshot. Therefore, the screenshot control unit **77** ignores the execution instruction for the screenshot and does not execute the screenshot (step S704). In step S704, the screenshot control unit **77** may disable the screenshot image of the display **20** (replacing all pixel values with a constant pixel value) or save an image that has been masked (replaced with a mask composed of a constant pixel value) in the non-volatile memory unit **62**.

[0119] Meanwhile, in the portable information terminal **1** described above, the conditions for executing code reading are controlled based on the distance to the workpiece **W** measured by the distance measuring sensor **37**. FIG. 7A is a flowchart showing an example of code reading condition control, and FIGS. 7B, 7C, and 7D are diagrams schematically showing the operations performed in the code reading condition control of FIG. 7A.

[0120] In step S801, the service unit **88** measures the distance to the workpiece **W** using the distance measuring sensor **37** and acquires the measured distance from the distance measuring sensor **37**. In step S802, the service unit **88** determines the position and size of the target range **J** to be displayed in the image view window **93** from the code image **Ic** obtained by the rear camera **33** capturing the camera field of view **V33**, according to the measured distance. This is to correspond to the fact that the distance to the workpiece **W** affects the spacing between the center **M33** of the camera field of view **V33** (in other words, the center of the code image **Ic**) and the aimer light **L36**, as well as the size of the code **C**, which changes in the code image **Ic**.

[0121] In FIG. 7B, a case where the distance to the workpiece **W** is relatively close is shown, and in FIG. 7C, a case where the distance to the workpiece **W** is relatively far is shown. As can be seen from these comparisons, when the distance to the workpiece **W** is close (FIG. 7B), the interval Δn between the center **M33** of the camera field **V33** appearing in the code image **Ic** and the aimer light **L36** is wider than the interval Δf between the center **M33** of the camera field **V33** appearing in the code image **Ic** and the aimer light **L36** when the distance to the workpiece **W** is far (FIG. 7C). Furthermore, when the distance to the workpiece **W** is close (FIG. 7B), the size of the code **C** appearing in the code image **Ic** is larger than the size of the code **C** appearing in the code image **Ic** when the distance to the workpiece **W** is far (FIG. 7C).

[0122] Therefore, the service unit **88** biases the center **Mj** of the target range **J** to be displayed in the image view window **93** from the center **M33** of the camera field of view **V33** to the irradiation position of the aimer light **L36** among the code image **Ic**. For example, the service unit **88** sets the center **Mj** of the target range **J** between the center **M33** of the camera field of view **V33** and the irradiation position of the aimer light **L36**. In particular, the service unit **88** narrows the interval **R** between the center **Mj** of the target range **J** in the camera field of view **V33** and the center **M33** of the camera field of view **V33** as the measurement distance becomes longer. Furthermore, the service unit **88** reduces the size of the target range **J** in the camera field of view **V33** as the measurement distance becomes longer. As a result, the farther the measurement distance, the more the aimer light **L36** and the code **C** are enlarged. Then, in step S803 (FIG. 7A), the service unit **88** displays the target range **J** determined in step S802 in the image view window **93**.

[0123] In step S804, the service unit **88** calculates the depth of field information **96** based on the measurement distance obtained in step S801 (FIG. 7D). The depth of field information **96** includes the position coordinate **961** up to infinity, the depth of field **962** at the position coordinate **961**, and the focus position **963** of the rear camera **33**. The depth of field **962** is calculated based on the measurement distance. The focus position **963** is adjusted to match the position of the workpiece **W** indicated by the measurement distance, as described above. The service unit **88** displays the depth of field information **96** shown in FIG. 7D in the image view window **93**. Furthermore, the service unit **88** compares the depth of field **962** with a predetermined depth threshold, and if the depth of field **962** is less than the depth threshold, a warning is displayed in the image view window **93**.

[0124] In step S805, the service unit **88** sends the measurement distance to the decoder **86**, and the

decoder **86** calculates the binning size corresponding to the measurement distance. Specifically, the decoder **86** calculates the binning size such that the closer the measurement distance, the larger the binning size. Here, binning refers to an image processing that combines adjacent pixels in the code image **Ic** into one pixel of a predetermined binning size, and the decoder **86** performs decoding processing on the code image **Ic** where binning has been executed.

[0125] And, until the display of the image view window **93** is ended (in step **S806** with “YES”), steps **S801** to **S805** are repeatedly executed.

[0126] The attachment **4** can be detachably attached to the housing **11** of the portable information terminal **1**. FIGS. **8A** to **8E** are rear views showing examples of the attachment, illustrating the state of the attachment **4** attached to the housing **11**. The attachment **4** has an attachment cover **41** that is attachable to the rear **13** of the housing **11**, and when the attachment cover **41** is attached to the housing **11**, it overlaps the rear side of the rear camera unit **32** in the thickness direction **Xt**.

[0127] The attachment cover **41** has windows **413**, **414**, **415**, **416**, and **417** that are opened. When the attachment **4** is attached to the housing **11**, the window **413** faces the rear camera **33**, the window **414** faces the wide-angle illumination unit **34**, the window **415** faces the narrow-angle illumination unit **35**, the window **416** faces the aimer light illumination unit **36**, and the window **417** faces the distance measuring sensor **37**. A pair of windows **417** is provided corresponding to the optical system that emits distance measuring light **L37** and the optical system that focuses the distance measuring light **L37** onto the optical sensor. Therefore, the rear camera **33** can capture the camera field of view **V33** through the window **413**, the wide-angle illumination unit **34** can emit wide-angle illumination light **L34** through the window **414**, the narrow-angle illumination unit **35** can irradiate narrow-angle illumination light **L35** through the window **415**, the aimer light illumination unit **36** can irradiate aimer light **L36** through the window **416**, and the distance measuring sensor **37** can irradiate and detect the distance measuring light **L37** through the window **417**.

[0128] In the attachment **4** shown in FIG. **8A**, a polarizing plate **423** is attached to the window **413** for the rear camera **33**, and a polarizing plate **425** is attached to the window **415** for the narrow-angle illumination unit **35**, with the polarization angles of the polarizing plate **423** and the polarizing plate **425** being orthogonal to each other. Therefore, the narrow-angle illumination light **L35** emitted from the narrow-angle illumination unit **35** is polarized by the polarizing plate **425** before being irradiated into the camera field **V33** of the rear camera **33**. Furthermore, the rear camera **33** captures the camera field **V33** using the light polarized by the polarizing plate **423**.

[0129] In the attachment **4** shown in FIG. **8B**, a light diffusion plate **43** is attached to the window **414** for the wide-angle illumination unit **34**. Therefore, the wide-angle illumination light **L34** emitted from the wide-angle illumination unit **34** is diffused by the light diffusion plate **43** before being irradiated onto the camera field **V33** of the rear camera **33**. This wide-angle illumination light **L34** functions as dome-shaped illumination light that is irradiated from all directions onto the camera field **V33**.

[0130] In the attachment **4** shown in FIG. **8C**, a prism mirror **44** is attached to a window **414** for the wide-angle illumination unit **34**. Therefore, the wide-angle illumination light **L34** emitted from the wide-angle illumination unit **34** is irradiated onto the camera field of view **V33** of the rear camera **33** via the prism mirror **44**. This prism mirror **44** has the function of reducing the angle at which the wide-angle illumination light **L34** is irradiated onto the camera field of view **V33**, or in other words, has the function of irradiating the wide-angle illumination light **L34** as low-angle illumination onto the camera field of view **V33**.

[0131] In the attachment **4** shown in FIG. **8D**, a color filter **45** is attached to the window **414** for the wide-angle illumination unit **34**. Therefore, the wide-angle illumination light **L34** emitted from the wide-angle illumination unit **34** is irradiated onto the camera field of view **V33** of the rear camera **33** through the color filter **45**. This color filter **45** has the function of allowing only light of a specific color (specific wavelength) among the wide-angle illumination light **L34** to pass through.

[0132] In this way, four types of attachments **4**, for example, shown in FIG. **8A** to FIG. **8E**, can be detachably attached to the housing **11**. In contrast, the portable information terminal **1** can determine the type of attachment **4** that is attached to the housing **11**. FIG. **9** is a block diagram showing the electrical configuration of the portable information terminal that determines the type of attachment.

[0133] As shown in FIG. **9**, the portable information terminal **1** has a short-range wireless communication unit **38** and a startup control unit **39**. The attachment cover **41** of the attachment **4** has an identifier built in that identifies the type of the attachment **4**, and the short-range wireless communication unit **38** reads the identifier of the attachment **4** via NFC (Near Field Communication) to determine the type of the attachment **4**. Furthermore, the startup control unit **39** controls the activation of the short-range wireless communication unit **38** according to the distance measured by the distance measuring sensor **37**.

[0134] That is, the startup control unit **39** detects the approach of the attachment **4** to the housing **11** based on the distance measured by the distance measuring sensor **37**. When the measured distance of the distance measuring sensor **37** is longer than the communicable distance by NFC, the startup control unit **39** puts the short-range wireless communication unit **38** to sleep. On the other hand, when the measured distance of the distance measuring sensor **37** becomes less than or equal to the communicable distance by NFC, the startup control unit **39** determines that the attachment **4** has approached the housing **11** and activates the short-range wireless communication unit **38**. Then, the activated short-range wireless communication unit **38** reads the identifier of the attachment **4** and determines the type of the attachment **4**.

[0135] In the above-described embodiment, a volatile memory unit **61** (first memory unit) and a non-volatile memory unit **62** (second memory unit) are provided as a memory unit capable of storing captured images (photo image Ip, code image Ic) (FIG. **2A**). Then, the volatile memory unit **61** is used for the decoding process among these. That is, the code image Ic generated by the rear camera **33** is stored in the volatile memory unit **61**, and the decoder **86** (reading unit) performs the decoding process based on the stored code image Ic. Thus, the code C (image processing target) is captured using the rear camera **33** of the portable information terminal **1**, and the decoding process is executed. In addition, a normal mode (first mode) and a security mode (second mode) are provided as modes for storing the code image Ic generated by the rear camera **33**. The normal mode stores the code image Ic generated by the rear camera **33** in the non-volatile memory unit **62** (step **S603** in FIG. **6D**). The security mode deletes the code image Ic generated by the rear camera **33** from the volatile memory unit **61** without storing it in the non-volatile memory unit **62** (step **S604** in FIG. **6D**). In the security mode, instead of this operation, a step of forcibly deleting the code image Ic stored in the non-volatile memory unit **62** from the non-volatile memory unit **62** may be executed. With this configuration, by selecting the security mode, the code image Ic generated by the rear camera **33** is deleted, so there is no leakage to the outside. Thus, when using the rear camera **33** mounted on the portable information terminal **1** for capturing the code C, the external leakage of the code image Ic captured by the rear camera **33** can be prevented.

[0136] Furthermore, a display **20** is further provided, and the service unit **88** (control unit) displays the code image Ic, which is stored in the volatile memory unit **61** and is the target of the decoding process of the decoder **86**, on the display **20** (image view window **93**) regardless of whether the normal mode of the security setting unit **74** (mode selection unit) is selected or the security mode is selected. With such a configuration, a user operating the portable information terminal **1** can perform tasks while visually recognizing the code image Ic, which is the target of the decoding process, on the display **20**, thereby improving the user's operability.

[0137] Furthermore, the screenshot prohibition unit **76** (control unit) prohibits the screenshot of the display **20** displaying the code image Ic when selecting the security mode of the security setting unit **74**, or stores the disabled or masked image of the screenshot of the display **20** displaying the code image Ic in the non-volatile memory unit **62** (FIG. **6E**). With this configuration, it is possible

to prevent the screenshot image from leaking to the outside.

[0138] Furthermore, a photo shooting processing unit **71** is provided that generates a photo image **Ip** using the rear camera **33** and stores it in the non-volatile memory unit **62**. The control unit **5** controls the rear camera **33** via either a camera control public API **51** (public API) that generates the photo image **Ip** in response to a request from the photo shooting processing unit **71**, or a camera control private API **52** (private API) that generates a code image **Ic** in response to a request from the code reading processing unit **72** (reading unit). In other words, in normal mode, both the control of the rear camera **33** via the camera control public API **51** and the control of the rear camera **33** via the camera control private API **52** are permitted. In security mode, while the control of the rear camera **33** via the camera control private API **52** is permitted, the control of the rear camera **33** via the camera control public API **51** is prohibited. In this configuration, the APIs (public API and private API) for photo shooting and decoding processing are separated, allowing for easy switching of security for photos taken by the rear camera **33**.

[0139] In addition, a telephone communication unit **83** that performs communication via a telephone line is provided. The communication control unit **81** (control unit) permits voice calls via the telephone communication unit **83** with telephone terminals of registered telephone numbers when selecting the security mode of the security setting unit **74**, while prohibiting voice calls via the telephone communication unit **83** with telephone terminals of unregistered telephone numbers. With this configuration, the range of telephone use can be restricted to prevent information from leaking externally.

[0140] Furthermore, the communication control unit **81** prohibits the recording of voice calls by the telephone communication unit **83** when selecting the security mode of the security setting unit **74**. With this configuration, it is possible to prevent the recording of voice calls from leaking to the outside.

[0141] An internet communication unit **82** that performs communication via the internet is provided. The communication control unit **81** prohibits the connection to the internet by the internet communication unit **82** when selecting the security mode of the security setting unit **74**. With this configuration, it is possible to prevent information leakage to the outside via the internet.

[0142] This embodiment can address the following issues. In other words, there is a need to easily perform reading of codes as described above and image processing such as OCR (Optical Character Recognition/Reader) using portable information terminals represented by smartphones. In response to this, by using the camera mounted on the portable information terminal for capturing images of the image processing target, it can meet such needs. In particular, by displaying the camera's field of view on the display unit of the smartphone, the user can confirm that the user is capturing the image processing target within the camera's field of view while performing operations.

[0143] However, there were cases where displaying the camera's field of view on the display unit impaired the user's operability. In other words, the display unit of the camera shows the user application screen used for user operations. This user application screen is provided with predetermined input items, and the results of image processing are input into the input item that has been focused on (in other words, selected) by the user. At this time, due to the display of the camera's field of view, the display on the display unit switches from the user application screen to the camera's field of view, which caused the user to lose sight of the input item the user was operating on.

[0144] In contrast, this embodiment allows the camera's field of view to be displayed while ensuring the user's operability with respect to the user application screen.

[0145] The portable information terminal according to this embodiment includes a camera that captures a workpiece to generate a captured image, an image processing unit that executes image processing based on the captured image, a service unit that sends the result of the image processing to a user application that has at least one input item and reflects the result of the image processing in the input item, and a display unit that displays the user application screen when the user

application is executed. The service unit, during the execution of the user application, when the camera is activated to perform image processing, maintains the state in which the input item focused on (in other words, selected) as the input target in the user application screen is visually recognizable by the user, and displays an image view window showing at least a part of the camera's field of view along with the user application screen on the display unit.

[0146] The portable information terminal control method according to this embodiment includes a step of capturing a workpiece using a camera mounted on the portable information terminal to generate a captured image, a step of executing image processing based on the captured image, a step in which a service unit sends the result of the image processing to a user application that has at least one input item and reflects the result of the image processing in the input item, and a step in which a display unit displays the user application screen when the user application is executed. The service unit, during the execution of the user application, when the camera is activated to perform image processing, maintains the state in which the input item focused on (in other words, selected) as the input target in the user application screen is visible to the user, and displays an image view window showing at least a part of the camera's field of view along with the user application screen on the display unit.

[0147] In this embodiment, when the rear camera **33** (camera) is activated to perform the decoding process (image processing), an image view window **93** showing at least a part of the camera field of view **V33** of the rear camera **33** is displayed on the display **20** (display unit) while maintaining a state in which the input item focused on as the input target among the input items of the user application screen **91** is visually recognizable by the user, along with the user application screen **91** (FIG. 3A). Thus, it becomes possible to display the camera field of view **V33** of the rear camera **33** while ensuring the operability of the user regarding the user application screen **91**.

[0148] Furthermore, the service unit **88** displays the image view window **93** in a semi-transparent manner over the user application screen **91**, and the area of the user application screen **91** that overlaps with the image view window **93** is displayed semi-transparently. With this configuration, the user can visually recognize the user application screen **91** displayed semi-transparently through the image view window **93**. In this way, it becomes possible to display the camera field of view **V33** of the rear camera **33** while ensuring the operability of the user regarding the user application screen **91**.

[0149] Furthermore, the service unit **88** increases the transparency of the image view window **93** as the saturation of the image displayed in the image view window **93** increases. In this configuration, the transparency of the image view window **93** (the degree to which the user application screen **91** is transparent to the image view window **93**) can be optimized according to the saturation of the image view window **93**. As a result, it becomes possible to display the camera field **V33** of the rear camera **33** while ensuring the operability of the user regarding the user application screen **91**.

[0150] A trigger button **23** is provided that generates a trigger to execute the decode process (image processing) in the decoder **86** (image processing unit). When the trigger button **23** is operated, the service unit **88** displays the image view window **93** on the display **20**, and closes the image view window **93** on the display **20** after the completion of the decode process. With this configuration, the image view window **93** can be displayed at a timing corresponding to the operation of the trigger button **23**.

[0151] Furthermore, the service unit **88** causes the decoder **86** to perform the decoding process after waiting for a predetermined period of time after the trigger button **23** is operated, and when the decoding process is completed, it closes the image view window **93** on the display **20**. In this configuration, the decoding process is executed after waiting for a predetermined period of time from the operation of the trigger button **23**. As a result, there is no situation where the image view window **93** closes simultaneously with the completion of the decoding process before the user can confirm the code **C** that is the target of the decoding process, allowing the user to reliably confirm the code **C** that is the target of the decoding process through the image view window **93**.

[0152] Furthermore, the decoder **86** is capable of executing the step **S505** (multiple code processing) in FIG. **6C**, which performs decoding processing for each of the multiple codes **C** attached to the workpiece **W**, and the step **S406** (single code processing) in FIG. **6B**, which performs decoding processing for a single code **C** attached to the workpiece **W**. In this regard, the decoder **86** increases the size of the image view window **93** displayed on the display **20** during step **S505** (multiple code processing) compared to the size of the image view window **93** displayed on the display **20** during step **S406** (single code processing). With this configuration, it is possible to display the image view window **93** at an appropriate size according to the number of codes **C** subject to decoding processing.

[0153] Furthermore, the service unit **88** displays a part (target range **J**) of the code image **Ic** (captured image) in the image view window **93**. With this configuration, a part (target range **J**) of the code image **Ic** can be displayed enlarged in the image view window **93**. The service unit **88** displays only a part of the field of view **V33** of the rear camera **33** in the image view window **93**. The service unit **88** excludes the area other than the corresponding part of the field of view **V33** of the rear camera **33** from the target of image processing. This is to avoid confusion for the user, as the user would not know which code the user has read by reading a code that is not visible in the image view window **93**. As another example, the image sensor of the rear camera **33** partially outputs pixel signals from a part of the area, and the service unit **88** displays the captured image corresponding to that part of the area of the image sensor in the image view window **93**. By narrowing the readout area from the image sensor, the processing can be speeded up.

[0154] A distance measuring sensor **37** is provided, which measures the distance to the workpiece **W** based on the result of detecting the distance measuring light **L37** reflected by the workpiece **W** after irradiating the distance measuring light **L37** onto the workpiece **W**. The service unit **88** displays the target range **J** of the code image **Ic** in the image view window **93** and changes the size of the target range **J** according to the distance detected by the distance measuring sensor **37** (steps **S801** and **S802** in FIG. **7A**). With such a configuration, an appropriate target range **J** corresponding to the distance to the workpiece **W** can be displayed in the image view window **93**.

[0155] Furthermore, an aimer light illumination unit **36** that irradiates the workpiece **W** with an aimer light **L36**, which is visible light, and a distance measuring sensor **37** that measures the distance to the workpiece **W** based on the result of detecting the distance measuring light **L37** irradiated onto the workpiece **W** and reflected by the workpiece **W** are provided. The service unit **88** displays a target range **J** centered on a center **Mj** (target position) that is biased to a position corresponding to the irradiation position of the aimer light **L36** from the center **M33** (imaging center) of the code image **Ic** in the code image **Ic** in the image view window **93**. At this time, the service unit **88** changes the center **Mj** of the target range **J** according to the distance detected by the distance measuring sensor **37** (FIG. **7B**, FIG. **7C**). With such a configuration, it is possible to execute the decoding process focusing on the range biased to the irradiation position of the aimer light **L36**, in other words, the existence range of the image of the code **C** to be decoded.

[0156] The distance measuring sensor **37** is provided to measure the distance to the workpiece **W** based on the result of detecting the distance measuring light **L37** reflected by the workpiece **W** after irradiating the distance measuring light **L37** onto the workpiece **W**. The service unit **88** calculates the depth of field of the rear camera **33** from the distance to the workpiece **W** measured by the distance measuring sensor **37** and displays it in the image view window **93** (step **S804** in FIG. **7A**, FIG. **7D**). With this configuration, the user can confirm the suitability of the depth of field through the image view window **93**.

[0157] Furthermore, the service unit **88** displays a warning in the image view window **93** when the depth of field is below the threshold. With this configuration, the user can confirm that the depth of field is inappropriate through the image view window **93**.

[0158] Furthermore, this embodiment can address the following issues. In other words, there is a need to easily perform reading of codes as described above and image processing such as OCR

(Optical Character Recognition/Reader) using portable information terminals represented by smartphones. In response to this, by using the camera mounted on the portable information terminal for imaging the image processing target, it can meet such needs.

[0159] Such image processing is operated by sequentially registering the results of image processing to a predetermined user application for use. Conventionally, since the user selected the image processing results to be registered in the user application, it required effort for the user's selection. In particular, there were cases where the user could not determine the necessity of registration to the user application, making such selection difficult for the user.

[0160] In contrast, this embodiment allows the user to easily confirm the necessity of registering the results of image processing using the camera of the portable information terminal to the user application.

[0161] The portable information terminal according to this embodiment includes a camera that captures a workpiece and generates a captured image, an image processing unit that performs image processing based on the captured image, at least one input item, and a service unit that sends the result of the image processing to a user application that reflects the result of the image processing in the input item and receives a response from the user application, and a display unit that displays an image view window showing the captured image. The user application determines the necessity of registration of the result of the image processing to the user application based on the result of the image processing received from the service unit, sends the determination result of the necessity of registration to the service unit, and the service unit displays an image view window on the display unit that overlays the information indicating the determination result received from the user application on the captured image.

[0162] The image processing result registration method according to this embodiment includes a step in which a camera captures a workpiece and generates a captured image, a step in which image processing is executed based on the captured image, and a user application that has at least one input item and reflects the result of the image processing in the input item, wherein a service unit sends the result of the image processing to the user application and receives a response from the user application, and a step in which an image view window showing the captured image is displayed on a display unit, and the user application determines the necessity of registering the result of the image processing to the user application based on the result of the image processing received from the service unit, sends the determination result of the necessity of registration to the service unit, and the service unit displays an image view window on the display unit that shows information indicating the determination result received from the user application overlaid on the captured image.

[0163] In this embodiment, the user application **87** determines the necessity of registration of the reading data Dc, which is the result of the decoding process received from the service unit **88**, based on the reading data Dc, and sends the determination result of the necessity of registration to the service unit **88** (steps S102 to S103 in FIG. 5A). Then, the service unit **88** displays an image view window **93** on the display **20**, which overlays the marks Ma and Mb (information) indicating the determination result received from the user application **87** onto the code image Ic (FIG. 5C). Therefore, the user can easily confirm the necessity of registration of the reading data Dc (image processing result) based on the marks Ma and Mb displayed in the image view window **93**. In this way, it becomes possible for the user to easily confirm the necessity of registration of the result of the decoding process using the rear camera **33** of the portable information terminal **1** to the user application **87**.

[0164] Furthermore, the user application **87** has data registration conditions **951** that include at least one condition related to the reading data Dc that requires registration to the user application **87**, and determines the necessity of registration based on the data registration conditions **951** (step S202 in FIG. 5B, FIG. 5C). With such a configuration, the necessity of registration of the reading data Dc can be easily and automatically determined based on the data registration conditions **951**.

[0165] Data registration condition **951** includes conditions related to the type of code C that is the target of the decoding process, the character string included in code C that is the target of the decoding process, the numerical values included in code C that is the target of the decoding process, and the position of code C that is the target of the decoding process. With such a configuration, it is possible to determine the necessity of registering the reading data Dc easily and automatically, which is the result of the decoding process, with appropriate conditions corresponding to code C that is the target of the decoding process.

[0166] Furthermore, the service unit **88** displays the image view window **93** on the display **20**, in which the code C that is the target of the decoding process included in the code image Ic currently displayed in the image view window **93**, which is identified by the mark Ma as requiring registration to the user application **87**, based on the determination result received from the user application **87** (step **S104** in FIG. 5A, FIG. 5C). With this configuration, the user can easily confirm that registration of the reading data Dc of the identified code C is necessary based on the image view window **93**.

[0167] Furthermore, the service unit **88** displays an image view window **93** on the display **20** that distinguishes between the codes C that require registration, the codes C that do not require registration because they are already registered, and the codes C that are unregistered and do not require registration. With this configuration, the user can easily confirm the registration status and necessity of each code C, which is the target of the decoding process, based on the image view window **93**.

[0168] Furthermore, when the decoder **86** receives a flick operation (registration operation) by the user while the image view window **93**, in which the code C requiring registration is specified by the mark Ma, is displayed on the display **20**, it registers the reading data Dc of the code C requiring registration (steps **S105** to **S106** in FIG. 5A). With this configuration, the user can easily register the reading data Dc of the code C requiring registration in the user application **87** by performing the flick operation.

[0169] Furthermore, this embodiment can address the following issues. In other words, there is a need to easily read codes using portable information terminals represented by smartphones. However, the cameras and lighting mounted on such portable information terminals are primarily intended for capturing photographs of people and landscapes, and are not necessarily suitable for reading codes. As a result, it took time for focusing and exposure, making it difficult to achieve rapid code reading.

[0170] This embodiment enables rapid reading of codes using the camera of a portable information terminal used for photography.

[0171] The portable information terminal according to this embodiment includes a housing having a first main surface provided with a display unit and a second main surface different from the first main surface, a camera that captures a field of view to generate a captured image with a variable focus position, a wide-angle illumination unit that irradiates wide-angle illumination light into the field of view of the camera, a narrow-angle illumination unit that irradiates narrow-angle illumination light into the field of view of the camera at an angle narrower than that of the wide-angle illumination light, an aimer light irradiation unit that irradiates visible light as aimer light onto a workpiece, a distance measuring sensor that measures the distance to the workpiece based on the result of detecting the distance measuring light irradiated onto the workpiece and reflected by the workpiece, a focus adjustment unit that adjusts the focus position of the camera according to the distance measured by the distance measuring sensor, a decoder that executes the decoding process of the code attached to the workpiece based on the captured image of the workpiece, and an imaging control unit that controls the conditions under which the camera captures images. The camera, wide-angle illumination unit, narrow-angle illumination unit, aimer light irradiation unit, and distance measuring sensor are provided on the second main surface of the housing, and the imaging control unit generates a captured image for photography by causing the camera to capture

images while irradiating wide-angle illumination light from the wide-angle illumination unit for photography, while for the decoding process, the imaging control unit causes the aimer light irradiation unit to irradiate the aimer light onto the workpiece, and while irradiating narrow-angle illumination light from the narrow-angle illumination unit, the imaging control unit causes the focus adjustment unit to adjust the focus position of the camera based on the distance measured by the distance measuring sensor to generate a captured image for decoding processing in the camera. [0172] In this embodiment, a housing **11** having a front surface **12** (first main surface) and a rear surface **13** (second main surface) is provided, with a display **20** (display unit) arranged on the front surface **12** and a rear camera **33** and a wide-angle illumination unit **34** arranged on the rear surface **13**. The wide-angle illumination unit **34** irradiates wide-angle illumination light **L34** into the camera field **V33** of the rear camera **33** while the rear camera **33** captures the camera field **V33**, thereby generating a photo image **Ip** (captured image) (photo shooting). Furthermore, a narrow-angle illumination unit **35**, an aimer light illumination unit **36**, and a distance measuring sensor **37** are provided on the rear surface **13**. The narrow-angle illumination unit **35** irradiates narrow-angle illumination light **L35** with a narrower light distribution angle than the wide-angle illumination light **L34** into the camera field **V33** of the rear camera **33**, the aimer light illumination unit **36** irradiates aimer light **L36**, which is visible light, onto the workpiece **W**, and the distance measuring sensor **37** irradiates distance measuring light **L37** onto the workpiece **W** and measures the distance to the workpiece **W** based on the result of detecting the distance measuring light **L37** reflected by the workpiece **W**. In the decoding process, the following operations are performed. That is, the user holds the portable information terminal **1** over the workpiece **W** so that the aimer light **L36** from the aimer light illumination unit **36** is irradiated onto the workpiece **W**. As a result, the distance measuring light **L37** from the distance measuring sensor **37** and narrow-angle illumination light **L35** from narrow-angle illumination unit are irradiated onto workpiece **W**. Consequently, focus position of rear camera is rapidly adjusted according to measured distance by distance measuring sensor and captured with relatively short exposure time of narrow angle illuminating lights thus generating code image **Ic**. Then decoder performs decoding process based on code image **Ic** thus rapid reading of code **C** made possible using rear camera of portable information terminal used for photo shooting. It should be noted that rolling shutter method more prone to shake compared to global shutter method. To suppress influence of shake it necessary shorten exposure time of rear camera here since narrow angle illuminating lights have higher illuminance than wide angle illuminating lights exposure time can be shortened by using them.

[0173] Furthermore, the aimer light **L36** and the distance measuring light **L37** are included in the camera field of view **V33** of the rear camera **33**, and are also included in the range where the narrow-angle illumination light **L35** is emitted. The positions of the aimer light **L36** and the distance measuring light **L37** on the workpiece **W** are included in the depth of field of the rear camera **33**, whose focus position is adjusted by the non-public API control unit **56** (focus adjustment unit). In this configuration, as the user holds the portable information terminal **1** over the workpiece **W** so that the aimer light **L36** is emitted from the aimer light illumination unit **36** to the workpiece **W**, the distance measuring light **L37** is emitted within the irradiation range of the narrow-angle illumination light **L35** on the workpiece **W**, allowing the focus of the rear camera **33** to be adjusted to the irradiation range based on the distance measured by the distance measuring sensor **37**. As a result, the rear camera **33** can generate the code image **Ic** with the focus set on the code **C** illuminated by the narrow-angle illumination light **L35**.

[0174] Furthermore, the optical axes of the aimer light **L36**, the distance measuring light **L37**, the camera field of view **V33** of the rear camera **33**, and the narrow-angle illumination light **L35** are parallel to each other. With this configuration, regardless of the distance to the workpiece **W**, the positional relationship of the aimer light **L36**, the distance measuring light **L37**, the camera field of view **V33** of the rear camera **33**, and the narrow-angle illumination light **L35** can be uniformly maintained.

[0175] The distance K35 between the optical axis of the narrow-angle illumination light L35 and the rear camera 33 is less than or equal to the distance K34 between the optical axis of the wide-angle illumination unit 34 and the optical axis of the rear camera 33. In this configuration, the narrow-angle illumination unit 35 is arranged in close proximity to the rear camera 33, allowing the narrow-angle illumination light L35 to accurately illuminate the code C within the camera field of view V33 of the rear camera 33.

[0176] Furthermore, the rear camera 33 is arranged in a line with any two of the narrow-angle illumination unit 35, the aimer light illumination unit 36, and the distance measuring sensor 37 (FIG. 1E, FIGS. 11A to 11D). With such a configuration, the narrow-angle illumination light L35, the aimer light L36, and the distance measuring light L37 can be brought close together and accommodated within the camera field of view V33 of the rear camera 33.

[0177] An attachment 4 for a rear camera 33 that is detachably attached to the housing 11 is provided. The attachment 4 has a polarizing plate 423 (optical component) that adjusts the characteristics of light incident on the rear camera 33 while being attached to the housing 11 and facing the rear camera 33. With this configuration, imaging by the rear camera 33 can be performed using an appropriate polarizing plate 423 according to the type of code C.

[0178] An attachment 4 of a narrow-angle illumination unit 35, which is detachably attached to the housing 11, is provided. The attachment 4 has a polarizing plate 425 (optical component) that adjusts the characteristics of the narrow-angle illumination light L35 emitted from the narrow-angle illumination unit 35, facing the narrow-angle illumination unit 35 while being attached to the housing 11. With such a configuration, the narrow-angle illumination light L35 can be emitted using an appropriate polarizing plate 425 according to the type of code C and the like.

[0179] Furthermore, a short-range wireless communication unit 38 (communication unit) that performs short-range wireless communication and a startup control unit 39 that controls the sleep and activation of the short-range wireless communication unit 38 are provided. The attachment 4 is equipped with an identifier that identifies the attachment 4, and the startup control unit 39 determines whether the attachment 4 is attached to the housing 11 based on the distance detected by the distance measuring sensor 37. Moreover, when the startup control unit 39 determines that the attachment 4 is not attached, it puts the short-range wireless communication unit 38 to sleep, while when it determines that the attachment 4 is attached, it activates the short-range wireless communication unit 38. Then, when the short-range wireless communication unit 38 is activated, it recognizes the identifier of the attachment 4 by the short-range wireless communication unit 38. With this configuration, the identifier of the attachment 4 attached to the housing 11 can be recognized by the short-range wireless communication of the short-range wireless communication unit 38. Furthermore, based on the distance detected by the distance measuring sensor 37, the short-range wireless communication unit 38 can be activated to recognize the identifier only when the attachment 4 is attached to the housing 11, which makes it possible to reduce the power consumption of the short-range wireless communication unit 38.

[0180] The decoder 86 changes the execution mode of the decoding process for the code image Ic according to the distance detected by the distance measuring sensor 37 (FIG. 7A). This allows the decoding process for the code image Ic to be executed in an appropriate execution mode according to the distance to the workpiece W.

[0181] For example, the decoder 86 performs decoding processing on the result of executing binning, which combines adjacent pixels into one pixel of a predetermined size, on the code image Ic. At this time, the decoder 86 changes the size in the binning according to the distance detected by the distance measuring sensor 37 (step S806 in FIG. 7A). With such a configuration, binning of the code image Ic can be performed with an appropriate size according to the distance to the workpiece W.

[0182] The decoder 86 executes the decoding process for the target range J having a center Mj (target position) that is biased from the center M33 (imaging center) of the code image Ic to a

position corresponding to the irradiation position of the aimer light **L36** among the code images **Ic**. At this time, the service unit **88** changes the target range **J** according to the distance detected by the distance measuring sensor **37** (step **S802** in FIG. **7A**). With this configuration, the decoding process can be executed focusing on the range biased from the irradiation position of the aimer light **L36**, in other words, the existence range of the image of the code image **Ic**.

[0183] The decoder **86** performs decoding process on the target range **J** of the code image **Ic**. At this time, the service unit **88** changes the size of the target range **J** according to the distance detected by the distance measuring sensor **37** (step **S802** in FIG. **7A**). With this configuration, the decoding process can be performed focusing on the target range **J** according to the distance to the workpiece **W**, in other words, the existence range of the code image **Ic**.

[0184] A trigger button **23** is provided to generate a trigger for starting the generation of the code image **Ic** by the rear camera **33** and the execution of the decoding process on the code image **Ic** in the service unit **88** (imaging control unit). The housing **11** has a shape extending in the longitudinal direction **Xl**, and has a right side surface **14** (side surface) that connects the front surface **12** (first main surface) and the rear surface **13** (second main surface) and extends along the longitudinal direction **Xl**. The right side surface **14** has a recessed portion **141** that is recessed inward between the front surface **12** and the rear surface **13**. The trigger button **23** has a base portion **231** that is arranged in the recessed portion **141** and extends in the longitudinal direction, and a protrusion **232** that protrudes outward from the right side surface **14** from the base portion **231**. With this configuration, the user can visually or tactilely recognize the protrusion **232** of the trigger button **23** that protrudes from the recessed portion **141** and can reliably operate the trigger button **23**.

[0185] The trigger button **23** is biased upward from the center **M11** of the housing **11** in the longitudinal direction (FIG. **1C**). In this configuration, since the trigger button **23** is offset from the center **M11** of the housing **11**, which is easy for the user to grip, it can suppress the user from accidentally operating the trigger button **23** when picking up the portable information terminal **1**. Moreover, during reading, since the user's thumb and index finger tend to be positioned on the upper side of the housing **11**, it is easy to operate the trigger button **23** even when the portable information terminal **1** is held over the code **C** in a difficult-to-read location.

[0186] This disclosure is not limited to the above embodiments, and various modifications can be made to the aforementioned as long as they do not deviate from the spirit thereof. For example, the method for setting the security mode may be changed as shown in FIG. **10**. FIG. **10** is a flowchart showing a modified example of the method for setting the security mode. In the modified example of FIG. **10**, the security setting unit **74** (FIG. **2A**) determines whether the administrator has performed an operation to instruct the setting of the security mode on the administrator setting screen (step **S901**). If there is no instruction for setting the security mode (in the case of "NO" in step **S901**), the security setting unit **74** sets the normal mode in step **S904** (that is, does not set the security mode).

[0187] On the other hand, if there is a setting instruction for the security mode (in the case of "YES" in step **S901**), the security setting unit **74** determines whether the portable information terminal **1** is within a predetermined security area (step **S902**). Specifically, the security setting unit **74** acquires the position information of the portable information terminal **1** calculated by the position information calculation unit **78** and determines whether the position indicated by the position information of the portable information terminal **1** is included in the security area. If the position of the portable information terminal **1** is not included in the security area (in the case of "NO" in step **S902**), the security setting unit **74** sets the normal mode in step **S904**. On the other hand, if the position of the portable information terminal **1** is included in the security area (in the case of "YES" in step **S902**), the security setting unit **74** sets the security mode in step **S903**.

[0188] In this modified example, the mode selection signal generated by the security setting unit **74** indicates which of the normal mode and the security mode is selected based on the settings by the administrator authority and the position information of the portable information terminal **1**. In such

a configuration, for example, if the position information indicates a position within the security area, the security mode is selected, and if the position information indicates a position outside the security area, the normal mode is selected, allowing for such operations. In the above embodiments and modified examples, two modes, the normal mode and the security mode, have been described, but this is not limited to these. That is, the security mode may include multiple security modes according to the security level, as shown in Table 1 below (1 to 4 in the table below), but is not limited to the combinations in Table 1.

TABLE-US-00001										TABLE 1										Normal										Security										Security										Security										Mode										Mode 1										Mode 2																																																																					
Mode 3										Mode 4										Storage of photo image										◦ x x x x										Ip in non-volatile memory unit										62										Storage of code image										Ic										◦ ◦ x x x										in non-volatile memory unit										62										Voice Call Recording										◦ ◦ ◦ x x										Internet connection										◦ ◦ ◦ ◦ x									

[0189] Furthermore, the layout of the rear camera unit 32 may be appropriately changed. FIGS. 11A to 11D are rear views showing modified examples of the layout in the rear camera unit. In the example of FIG. 11A, in the short side direction Xs, the rear camera 33 is arranged between the aimer light illumination unit 36 and the distance measuring sensor 37, and in the long side direction Xl, the rear camera 33 is arranged between the wide-angle illumination unit 34 and the narrow-angle illumination unit 35. Furthermore, the distance K35 between the rear camera 33 and the narrow-angle illumination unit 35 is less than or equal to the distance K34 between the rear camera 33 and the wide-angle illumination unit 34.

[0190] In the example of FIG. 11B, in the short-side direction Xs, the rear camera 33 is arranged between the aimer light illumination unit 36 and the distance measuring sensor 37. The wide-angle illumination unit 34 and the narrow-angle illumination unit 35 are arranged below the rear camera 33. In the short-side direction Xs, the narrow-angle illumination unit 35 is arranged on one side of the rear camera 33, and the wide-angle illumination unit 34 is arranged on the other side of the rear camera 33. Furthermore, the distance K35 between the rear camera 33 and the narrow-angle illumination unit 35 is less than or equal to the distance K34 between the rear camera 33 and the wide-angle illumination unit 34.

[0191] In the example of FIG. 11C, in the short-side direction Xs, the wide-angle illumination unit 34, the narrow-angle illumination unit 35, the rear camera 33, the aimer light illumination unit 36, and the distance measuring sensor 37 are arranged in this order. That is, the narrow-angle illumination unit 35 is arranged between the rear camera 33 and the wide-angle illumination unit 34, and the aimer light illumination unit 36 is arranged between the rear camera 33 and the distance measuring sensor 37. Furthermore, the distance K35 between the rear camera 33 and the narrow-angle illumination unit 35 is less than or equal to the distance K34 between the rear camera 33 and the wide-angle illumination unit 34, and the distance K36 between the rear camera 33 and the aimer light illumination unit 36 is less than or equal to the distance K37 between the rear camera 33 and the distance measuring sensor 37.

[0192] In the example of FIG. 11D, in the short-side direction Xs, the wide-angle illumination unit 34, the rear camera 331, the aimer light illumination unit 36, the rear camera 33, the narrow-angle illumination unit 35, and the distance measuring sensor 37 are arranged in this order. That is, the aimer light illumination unit 36 is arranged between the rear camera 33 and the wide-angle illumination unit 34, and the rear camera 331 is further arranged between the aimer light illumination unit 36 and the wide-angle illumination unit 34. In addition, the narrow-angle illumination unit 35 is arranged between the rear camera 33 and the distance measuring sensor 37. The distance K35 between the rear camera 33 and the narrow-angle illumination unit 35 is less than or equal to the distance K34 between the rear camera 33 and the wide-angle illumination unit 34, and is also less than or equal to the distance K37 between the rear camera 33 and the distance measuring sensor 37. The distance K36 between the rear camera 33 and the aimer light illumination unit 36 is less than or equal to the distance K34 between the rear camera 33 and the wide-angle illumination unit 34.

[0193] The structural differences between the rear camera **33** and the rear camera **331** can be variously assumed. For example, the rear camera **331** may be configured to have a longer imaging distance than the rear camera **33**. Alternatively, the rear camera **331** may be configured to have a wider field of view angle than the field of view angle of the rear camera **33**.

[0194] Furthermore, the manner of displaying the image view window **93** along with the user application screen **91** can be appropriately changed. FIGS. **12A** and **12B** are diagrams showing a first modified example of the display manner of the user application screen and the image view window. FIG. **12A** shows the display of the display **20** before the image view window **93** is displayed, and FIG. **12B** shows the display of the display **20** after the image view window **93** is displayed. In this first modified example, the service unit **88** acquires position information indicating the position of the input item **A3** that is focused by the user's selection from the user application **87**. Then, the service unit **88** displays the image view window **93** in a range outside the position of the focused input item **A3** indicated by the position information on the display **20**.

[0195] That is, the service unit **88** receives the position information of the input item **A3**, which is focused as the input target, from the user application **87** on the user application screen **91**. Then, the service unit **88** displays the image view window **93** at a position that does not overlap with the focused input item **A3** on the user application screen based on the position information (FIG. **12B**). In this configuration, the image view window **93** is displayed so as not to overlap with the focused input item **A3** on the user application screen **91**, allowing the user to visually recognize the input item **A3**. Thus, it becomes possible to display the camera field of view **V33** of the rear camera **33** while ensuring the user's operability with respect to the user application screen **91**.

[0196] Alternatively, the image view window **93** can also be displayed in a display area specified by the user. FIGS. **13A** and **13B** are diagrams showing a second modified example of the display mode between the user application screen and the image view window. FIG. **13A** shows the display of the display **20** before the image view window **93** is displayed, and FIG. **13B** shows the display of the display **20** after the image view window **93** is displayed.

[0197] In this second modified example, the user can specify the display area of the image view window **93** on the display **20** by operating the display **20**. This specification of the display area can be performed by setting the display area to the service unit **88** through the operation of the display **20**, or by executing a predetermined operation (double tap or long touch) at the position on the display **20** that the user wants to specify as the display area. When the service unit **88** receives this specification of the display area, it displays the image view window **93** in the specified display area on the display **20**, and displays the user application screen **91** in the area outside the specified display area (FIG. **13B**). FIG. **13B** shows the case where the upper part of the display **20** is specified as the display area.

[0198] That is, the service unit **88** accepts the specification of the display area of the image view window **93** on the display **20** and displays the user application screen **91** outside the display area of the image view window **93** on the display **20**. In this configuration, the position of the image view window **93** is predetermined, and when displaying the image view window **93**, the user application screen **91** is displayed outside of it, allowing the user to visually recognize the focused input item **A3**. In this way, it becomes possible to display the camera field of view **V33** of the rear camera **33** while ensuring the user's operability with respect to the user application screen **91**.

[0199] Furthermore, the control of the display period of the image view window **93** may be changed as appropriate. FIG. **14A** is a flowchart showing a first modified example of the display control of the image view window. The service unit **88** confirms that the trigger button **23** has been pressed ("YES" in step **S1001**), displays the image view window **93** on the display **20** as described above (step **S1002**), and starts the decoding process in the decoder **86** (step **S1003**). Then, when the decoder **86** completes the decoding process (step **S1004**), the service unit **88** monitors whether the trigger button **23** has been pressed again (step **S1005**). When it confirms that the trigger button **23** has been pressed again ("YES" in step **S1005**), the service unit **88** ends the display of the image

view window **93** (step **S1006**).

[0200] That is, the service unit **88** executes the decoding process (image processing) in the decoder **86** (image processing unit) when the trigger button **23** is operated, and when the trigger button **23** is operated again after the completion of the decoding process, it closes the image view window **93** on the display **20**. In this configuration, the image view window **93** displayed by the operation of the trigger button **23** is continuously displayed until the trigger button **23** is operated again. In other words, there is no situation where the image view window **93** closes simultaneously with the completion of the decoding process before the user has time to confirm the code C, which is the target of the decoding process, and the user can reliably confirm the code C, which is the target of the decoding process, through the image view window **93**.

[0201] FIG. **14B** is a flowchart showing a second modified example of display control of the image view window. In this second modified example, when the user application **87** focuses on the input item by a designated operation on the user's display **20**, it notifies the service unit **88** of the focus of the input item ("YES" in step **S1101**), it displays the image view window **93** on the display **20** (step **S1102**).

[0202] That is, the user application **87** notifies the service unit **88** that the input item has been specified via the display **20**, and the service unit **88**, upon receiving the notification from the user application **87**, displays the image view window **93** on the display **20**. With such a configuration, the user can quickly display the image view window **93** at the timing when the user wants to perform an operation related to the decoding process (that is, the timing when the input item is specified).

[0203] Furthermore, the registration management of the reading data Dc by the user application **87** may be configured as shown in FIG. **15**. FIG. **15** is a flowchart showing an example of the registration management of the reading data by the user application. In step **S1201**, the user application **87** checks the difference between the number of registered reading data Dc (number of registrations) registered in the data management table **95** and the specified number. Then, in step **S1202**, the user application **87** determines whether there is a shortage number of registrations against the specified number (in other words, it determines whether the number obtained by subtracting the number of registrations from the specified number is one or more).

[0204] If a shortage exists (when "YES" in step **S1202**), the user application **87** sends the corresponding shortage to the service unit **88** (step **S1203**), and the service unit **88** displays the shortage received from the user application **87** on the display **20** (for example, the image view window **93**). Furthermore, the combination format for sending multiple reading data Dc to the user application **87** later is sent from the user application **87** to the service unit **88** (step **S1204**). Not limited to this, the user may define the format in the configuration file of the terminal prepared by the user. This combination format indicates, for example, that multiple reading data Dc are combined by separating them with commas. Therefore, when the service unit **88** confirms the registration instruction operation in the multiple code reading mode of FIG. **5A** (step **S105**), it combines the multiple reading data Dc subject to the registration instruction operation according to the combination format and sends it to the user application **87**. If no shortage exists (when "NO" in step **S1202**), the user application **87** sends the specified number of reading data Dc to the external host computer (step **S1205**).

[0205] That is, the user application **87** specifies a combination format to the service unit based on the registration status of the user application **87** and the result of the reading data Dc sent from the service unit **88**, which will be used to combine the data sent in subsequent registration operations (steps **S1201**, **S1204**). Then, the service unit **88** combines the reading data Dc, which is the result of the decoding process of multiple codes C included in the image view window **93**, based on the specified combination format and sends it to the user application **87**. In this configuration, the reading data Dc, which is the result of the decoding process of multiple codes C, can be combined in an appropriate combination format and sent to the user application **87** in a consolidated manner.

[0206] Furthermore, the user application **87** sends the accumulated number of registered reading data Dc (the result of the decoding process) sent from the service unit **88** when it reaches the specified number (when “NO” in step **S1202**) to the external host (in step **S1205**). With this configuration, the frequency of sending from the user application **87** to the external host can be suppressed, thereby reducing the load associated with these.

[0207] Furthermore, the user application **87** sends the shortage number up to the specified number to the service unit **88** based on the number of registered reading data Dc and the specified number (step **S1203**), and the service unit **88** displays the shortage number on the display **20**. This configuration allows the user to easily check the shortage number via the display **20**. Additionally, not only when sent from the user application **87**, the service unit **88** can also recognize the registration progress status of the image processing results based on the response from the user application **87**. If the service unit **88** has previously obtained the information of the specified number, it can also recognize the shortage number. By displaying the registration progress status on the display **20**, the user can easily grasp the current number of reading data and the shortage number. The service unit **88** may obtain the judgment conditions regarding the timing for sending the registered image processing results from the user application **87** to the external host. The judgment conditions may include, for example, the number of reading data registered in the user application (when 10 are registered), the code type (when a specific code type is read), user operations (when double-tapped), and so on, but are not limited to these. The judgment conditions may be defined not only in the format information sent from the user application **87** to the service unit **88** but also in the configuration file of the terminal prepared by the user. When the judgment conditions are satisfied, the service unit **88** sends an instruction to the user application **87** to send the registered image processing results from the user application **87** to external host. This allows the user application side to eliminate the trouble of implementing the logic to determine the timing for sending to external host.

[0208] Furthermore, the specific manner of the administrator's security mode setting instruction for the security setting unit **74** can be changed as appropriate. In other words, in the above-mentioned embodiment, the security setting unit **74** displays the administrator setting screen on the display **20** and accepts the administrator's security mode setting instruction based on the operation on the administrator setting screen by the administrator. In contrast, in the modified example described here, the security setting unit **74** is equipped with a communication function with an external management server used by the administrator. The security setting unit **74** receives a mode selection signal from the management server.

[0209] That is, the security setting unit **74** (communication unit) has a communication function that receives a mode selection signal. Then, the security setting unit **74** selects either the security mode or the normal mode based on the mode selection signal received by the communication function. Here, the mode selection signal indicates which of the security mode and the normal mode is selected based on the setting by administrator authority. With such a configuration, it is possible to prevent users other than the administrator and finders from switching between the security mode and the normal mode.

[0210] Furthermore, in the above example, during code reading, the narrow-angle illumination unit **35** irradiates the narrow-angle illumination light L35 in the camera field of view V33, and the rear camera **33** captures the code C within the camera field of view V33 to obtain the code image Ic. However, the wide-angle illumination unit **34** may also be used during code reading. For example, it is anticipated that overexposure may occur in the code image Ic captured using the narrow-angle illumination light L35, resulting in a failure to read the reading data Dc during the decoding process of the code image Ic. In such a case, the illumination unit used may be switched from the narrow-angle illumination unit **35** to the wide-angle illumination unit **34**, and the code reading may be re-executed. Specifically, the wide-angle illumination unit **34** irradiates the wide-angle illumination light L34 in the camera field of view V33, and the rear camera **33** captures the code C

within the camera field of view V33 to obtain the code image Ic. Then, the decoder 86 executes the decoding process on the code image Ic to obtain the reading data Dc.

[0211] That is, the service unit 88 (imaging control unit) adjusts the focus position of the rear camera 33 by the non-public API control unit 56 (focus adjustment unit) while irradiating the narrow-angle illumination unit 35 with narrow-angle illumination light L35, and if the decoding process for the generated code image Ic captured by the rear camera 33 fails, it executes the decoding process for the generated code image Ic by adjusting the focus position of the rear camera 33 by the non-public API control unit 56 (focus adjustment unit) while irradiating the wide-angle illumination unit 34 with wide-angle illumination light L34. In such a configuration, if the decoding process using the narrow-angle illumination light L35 fails due to overexposure or the like, there is a possibility of success in the decoding process using the wide-angle illumination light L34.

[0212] In the example of FIG. 2A, in order to prohibit photo shooting in security mode, a camera control public API 51 and a camera control private API 52 are provided, and these are used selectively. However, it may also be configured to prepare only the camera control private API 52 and allow access to the camera module 3 only from specific applications, such as the code reading processing unit 72, during security mode.

[0213] Furthermore, as a specific example of image processing, it is not limited to the above-mentioned decoding process, but also includes other image processing such as OCR.

[0214] This invention is applicable to the technology for reading codes attached to a workpiece in general.

Claims

1. A portable information terminal comprising: a camera that captures a workpiece and generates a captured image, a reading unit that executes image processing based on the captured image to acquire information contained in the workpiece, a first memory unit that is a volatile memory unit and temporarily stores the captured image, a second memory unit that is a non-volatile memory unit and stores settings of the image processing and is capable of storing the captured image, a control unit that causes the camera to generate the captured image and store the captured image in the first memory unit, and causes the reading unit to execute the image processing based on the captured image in the first memory, a mode selection unit that selects any of a plurality of modes including a first mode where the captured image is stored in the second memory unit and a second mode where the captured image is deleted from the first memory unit without the captured image being not stored in the second memory unit or the captured image is forcibly deleted from the second memory unit after being stored in the second memory.
2. The portable information terminal according to claim 1, further comprising: a communication unit that receives a mode selection signal, wherein the mode selection unit selects one of the plurality of modes based on the mode selection signal received via the communication unit, and the mode selection signal indicates which of the plurality of modes is selected based on settings by an administrator authority.
3. The portable information terminal according to claim 2, wherein the mode selection signal indicates which of the plurality of modes is selected based on the setting by the administrator authority and the position information of the portable information terminal.
4. The portable information terminal according to claim 1, further comprising: a display unit, wherein the control unit displays, on the display unit, the captured image stored in the first memory unit and subject to the image processing by the reading unit, regardless of whether the first mode or the second mode is selected by the mode selection unit.
5. The portable information terminal according to claim 4, wherein the control unit prohibits a screenshot of the display unit displaying the captured image when the second mode is selected, or

stores an image that is either disabled or masked of the screenshot of the display unit in the second memory unit.

6. The portable information terminal according to claim 1, further comprising: a photo shooting processing unit that generates the captured image by the camera and stores the captured image in the second memory unit, wherein the control unit controls the camera via either a public API that controls the camera to generate the captured image in response to a request from the photo shooting processing unit, or a private API that controls the camera to generate the captured image in response to a request from the reading unit, in the first mode, both the control of the camera via the public API and the control of the camera via the private API are permitted, and in the second mode, while the control of the camera via the private API is permitted, the control of the camera via the public API is prohibited.

7. The portable information terminal according to claim 1, further comprising: a telephone communication unit that performs communication via a communication line, wherein in the second mode, the control unit allows voice calls via the telephone communication unit with telephone terminals of registered telephone numbers, while prohibiting the voice calls via the telephone communication unit with telephone terminals of unregistered telephone numbers.

8. The portable information terminal according to claim 7, wherein the control unit prohibits recording of voice calls by the telephone communication unit in the second mode.

9. The portable information terminal according to claim 1, further comprising: an internet communication unit that performs communication via internet, wherein the control unit prohibits connection to the internet by the internet communication unit in the second mode.

10. The portable information terminal according to claim 1, wherein the plurality of modes includes a third mode where the captured image, for which image processing is executed, is stored in the second memory unit, while a photo image generated by the camera is deleted from the first memory unit without being stored in the second memory unit, or the photo image is forcibly deleted from the second memory unit after being stored in the second memory unit.

11. An image management method for managing a first memory unit or a second memory unit of a captured image generated by a camera that captures a workpiece, comprising: a step of selecting any of a plurality of modes including a first mode where the captured image generated by the camera is stored in the second memory unit, and a second mode where the captured image is deleted from the first memory unit without being stored in the second memory unit, or the captured image is forcibly deleted from the second memory unit after being stored in the second memory unit; and a step of controlling the camera to generate the captured image and storing the captured image in the first memory unit, and executing image processing based on the captured image in the first memory unit to acquire information included in the workpiece, wherein the first memory unit is a volatile memory unit that temporarily stores the captured image, and the second memory unit is a non-volatile memory unit that is capable of storing the captured image and stores settings of the image processing.
